Posterior Variance Collapse in Conditional Flow Matching is the Zero-Bandwidth Limit of a Kernel-Weighted Empirical Conditional Flow

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Abstract

Conditional flow matching (CFM) trained to convergence on a finite dataset is often described as "memorising" the training data: overtraining is reported to collapse the conditional generative law onto individual training examples. We give an exact, population-level account of why this happens. Fixing the dataset and the deterministic linear interpolant, the L^2-optimal velocity field is a posterior-weighted average of per-example collapsed fields, and under hard conditioning with identifying labels this average degenerates to a single atom: the exact minimiser has *zero* conditional variance and zero loss (Proposition 2). We then show that the standard remedy—label smoothing, i.e. adding Gaussian noise to the conditioning variable—turns this hard selection into *exact kernel regression over the training atoms*: the generated endpoint law becomes a Nadaraya–Watson mixture $\sum_i p_i^{(h)}(y)\,\delta_{x^i}$ (Theorem 1) whose moments are a parameter-free reference curve computable directly from the training set (Proposition 5). This remedy, however, can only match moments; it cannot recover the posterior, since the generated law remains atomic at every bandwidth h, yielding an h-independent Wasserstein floor (Proposition 6). We further show that two superficially similar remedies are not interchangeable: label noise moves the population endpoint, while interpolant noise provably does not, at any noise level (Proposition 9). Across five seeds, a trained MLP tracks the kernel reference to within seed noise (ratio-to-kernel ≈ 1.00 for $h \geq 0.05$), and we reproduce the same collapse on a Gaussian-mixture posterior, on MNIST and CIFAR-10 inpainting, under adversarially shuffled labels that carry no real information about x, and across a range of architectures and optimisers. Throughout, we distinguish carefully between exact population consequences and empirical findings that the population theory does not, by itself, determine.

1 Introduction

Flow matching and stochastic-interpolant models are usually trained by regressing a velocity field against a linear path between a noise sample and a data sample. On a finite dataset, exact minimisation of this objective against the *empirical* data law is known to induce memorisation: deployed diffusion models can be made to reproduce individual training images verbatim (Carlini et al., 2023; Somepalli et al., 2023), and a recent line of work has documented that conditional flow-matching models, trained past the point of good posterior calibration, produce increasingly deterministic (low-variance) conditional samples that coincide with individual training examples (Dasgupta et al., 2026; Reu et al., 2025).

We give an exact, population-level account of this mechanism. Fixing the training set $\mathcal{D} = \{(x^i, y^i)\}_{i=1}^N$ and treating only the interpolation randomness as random, we characterise the exact population minimiser of the flow-matching objective over all square-integrable vector fields, together with the endpoint law of its flow.

The observation that the empirical minimiser memorises is *not* new; it is well established in the diffusion-memorisation literature (Pidstrigach, 2022; Gu et al., 2023; Kadkhodaie et al., 2024; Biroli and Mézard, 2023; Kamb and Ganguli, 2024; Buchanan et al., 2025). Closest in spirit is Buchanan et al. (2025), who independently characterise a memorisation/generalisation *phase transition* in *unconditional* diffusion models by comparing the training loss of a Gaussian-mixture denoiser at capacity M to that of a memorising surrogate, as M ranges from the true number of modes K up to the dataset size N, and show that the resulting crossover point $M_\star$ is linear in N. This is a capacity-driven account of the same broad phenomenon we study from the label/bandwidth side: their $M\colon K \to N$ interpolation is the model-capacity analogue of our $h\colon 0 \to \infty$ interpolation (Corollaries 2 and 3), and neither result subsumes the other.

Our contribution is the *conditional and kernel-regression structure* underlying this collapse, which the existing literature does not isolate:

1. **Conditional sharpening.** Hard conditioning turns full-empirical-measure memorisation into *single-atom* memorisation—zero conditional variance (Corollary 1). The correct contrast with the unconditional model is not "memorises vs. does not" but *which* empirical measure is memorised.
2. **Label smoothing $\equiv$ kernel regression.** Adding label noise makes the population endpoint an exact Nadaraya–Watson mixture over training atoms (Theorem 1), converting a $0/1$ statement into a quantitative, parameter-free reference curve $\mathrm{Cov}_h(y)$ (Proposition 5).
3. **Separation of two remedies.** Label noise changes the population endpoint; interpolant noise provably does not, for any noise level (Propositions 9 and 10).
4. **Atomicity obstruction.** The smoothed law is atomic for every h, yielding an h-independent Wasserstein floor (Proposition 6): matching a second moment is not recovering the posterior.
5. **Representation floor.** A uniform Lipschitz bound on the velocity field gives a measurable loss floor $d/(3L)$ (Corollary 5) that separates representation error from optimisation error.
6. **Finite-sample vs. population representability.** To close a gap between the fixed-dataset population theory above and finite-sample training, we adapt two representability results of the unconditional gradient-variance/ stochastic-interpolant literature (Reu et al., 2025) into the present notation and setting (Proposition 12, Remark 1): a fixed finite unconditional batch always admits an exact zero-loss memorising field despite positive population loss, and training-time resampling of x_0 is exactly what defeats this construction. Building on this, we give a new, problem-specific exact-representability result: no finite-depth, finite-width network with globally Lipschitz activations can equal the collapsed field's $1/(1-t)$ blow-up *pointwise* on all of $[0, 1)$ (Lemma 3). We keep this pointwise claim explicitly separate from a strict loss floor: unbounded pointwise error on a shrinking neighbourhood of $t = 1$ does not by itself imply $L_0(v_\theta)$ stays bounded away from 0, so the only class for which we prove a genuine positive loss floor is the Lipschitz-capped one of Corollary 5, not an arbitrary finite architecture with unbounded weights.

In one line: *conditional collapse is the zero-bandwidth limit of a kernel-weighted empirical conditional flow.*

A word on scope, developed formally in Section 3.6. Everything in the theory characterises the exact population minimiser for the empirical data law; it is not a statement about what a finite network trained by SGD attains. We therefore distinguish sharply between quantities the population theory determines (and which our experiments confirm) and quantities it does not (dependence of observed collapse on dataset size or observation noise), which we report as empirical findings rather than "predictions."

2 Setup and Notation

Fix a training set $\mathcal{D} = \{(x^i, y^i)\}_{i=1}^N$ with $x^i \in \mathbb{R}^d$, $y^i \in \mathbb{R}^k$, empirical joint law $\widehat{\rho}_{XY} = \frac{1}{N}\sum_i \delta_{(x^i,y^i)}$ and marginal $\widehat{\rho}_X = \frac{1}{N}\sum_i \delta_{x^i}$. The training set is fixed (deterministic); randomness comes only from

$$I \sim \mathrm{Unif}\{1, \dots, N\}, \qquad X_0 \sim \pi_0 = \mathcal{N}(0, I_d), \qquad t \sim \mathrm{Unif}(0, 1),$$

drawn independently, together with label noise $\varepsilon \sim \mathcal{N}(0, I_k)$ when $h > 0$. Set $X_1 = x^I$ and, for the deterministic linear interpolant,

$$X_t = (1-t)X_0 + tX_1, \qquad U := \dot{X}_t = X_1 - X_0.$$

The smoothed label is $\widetilde{Y} = y^I + h\varepsilon$ with Gaussian kernel $K_h(u) = (2\pi h^2)^{-k/2} \exp(-|u|^2/2h^2)$. The objective is

$$L_h(v) = \mathbb{E}\big[\,|U - v(X_t, t, \widetilde{Y})|^2\,\big]. \tag{1}$$

For $h = 0$ we write $\widetilde{Y} = Y = y^I$ and L_0. Let $\bar{x} = \frac{1}{N}\sum_i x^i$ and $\widehat{\Sigma}_X = \frac{1}{N}\sum_i (x^i - \bar{x})(x^i - \bar{x})^\top$. In the linear-Gaussian instance used experimentally, $y = Ax + \eta$ with $\eta \sim \mathcal{N}(0, \sigma^2_{\text{obs}} I_k)$, which yields a Gaussian analytic posterior with covariance Σ_{post}.

Scope marker. Everything in Sections 3.1–3.3 characterises the exact population minimiser over L^2-measurable vector fields for the empirical data law. Section 3.4 quantifies one reason a finite trained network differs.

3 Theory

We state the load-bearing results with proofs; auxiliary lemmas are proved in Appendix A. The projection identity underlying everything is standard.

Proposition 1 (L^2 projection)**.** *For every square-integrable v, $L_h(v) = \mathbb{E}[\mathrm{Var}(U \mid X_t, t, \widetilde{Y})] + \mathbb{E}\,|v - \mathbb{E}[U \mid X_t, t, \widetilde{Y}]|^2$, so the a.e.-unique minimiser is $v^\star_h(x, t, y) = \mathbb{E}[U \mid X_t = x, t, \widetilde{Y} = y]$ with $\inf_v L_h = \mathbb{E}[\mathrm{Var}(U \mid X_t, t, \widetilde{Y})]$.*

Two facts make the conditional expectations legitimate and the flows well posed without any additional assumption (Appendix A): (i) for $t < 1$ the map $X_0 \mapsto X_t = (1-t)X_0 + tx^i$ is an invertible affine change of variables, with $X_0 = (X_t - tx^i)/(1-t)$ and conditional density $p(x \mid I = i, t) = (1-t)^{-d}\pi_0\big(\frac{x - tx^i}{1-t}\big) > 0$ for every x (Lemma 4); (ii) any field of the form $v(x,t) = \sum_i w_i(x,t)\,(x^i - x)/(1-t)$ with smooth weights summing to one is C^∞ hence locally Lipschitz on $\mathbb{R}^d \times [0,1)$, so its ODE has a unique solution up to $t = 1$ and endpoint laws are the well-defined weak limits $p_1 = \lim_{t\uparrow 1} p_t$ (Lemma 5).

3.1 Hard conditioning: exact collapse

We begin with the base case: a raw, unsmoothed label ($h = 0$) that identifies its training example uniquely. Proposition 1 already tells us the exact minimiser is a conditional expectation; the question is what that conditional expectation collapses to once the conditioning event pins down a single index i.

Proposition 2 (Exact conditional collapse; $h = 0$)**.** *Assume $y^1, \ldots, y^N$ are pairwise distinct. Then (a) $v^\star_0(x, t, y^i) = (x^i - x)/(1-t)$ for $t < 1$; (b) $\inf_v L_0(v) = 0$; (c) the ODE $\dot{x}_t = (x^i - x_t)/(1-t)$ has the explicit solution $x_t = (1-t)x_0 + tx^i$, so $x_t \to x^i$ for every x_0; and (d) $p_1^{\text{cond}}(\cdot \mid y^i) = \delta_{x^i}$, hence $\mathrm{Cov}[p_1^{\text{cond}}(\cdot \mid y^i)] = 0$.*

Proof. (a) Distinctness makes $\{Y = y^i\}$ identify $I = i$, so $X_1 = x^i$ is deterministic and, by Lemma 4, $X_0 = (x - tx^i)/(1-t)$ is determined by (x, t, i). Hence $U = x^i - X_0 = (x^i - x)/(1-t)$ is $\sigma(X_t, t, Y)$-measurable and equals its conditional expectation. (b) $U - v^\star_0 = 0$ a.s. (c) Multiplying by $(1-t)$ gives $\frac{d}{dt}\big[\frac{x_t - x^i}{1-t}\big] = 0$, so $\frac{x_t - x^i}{1-t} \equiv x_0 - x^i$. (d) The endpoint is x^i independently of x_0. $\square$

If labels are not distinct, collapse is onto the label's empirical conditional support: for $\mathcal{I}_y = \{i : y^i = y\}$, $p_1^{\text{cond}}(\cdot \mid y) = |\mathcal{I}_y|^{-1}\sum_{i\in\mathcal{I}_y}\delta_{x^i}$. Collapse is always to the empirical conditional law, which is a single atom exactly when the label is an identifier.

Proposition 3 (Unconditional endpoint is the full empirical law)**.** *For $L_{\text{unc}}(v) = \mathbb{E}|X_1 - X_0 - v(X_t, t)|^2$ and $t < 1$, with $q_i(x,t) = \Pr(I = i \mid X_t = x, t) \propto \pi_0\big(\frac{x - tx^i}{1-t}\big)$,*

$$v^\star_{\text{unc}}(x, t) = \sum_i q_i(x,t)\,\frac{x^i - x}{1-t}, \qquad p_1^{\text{unc}} = \widehat{\rho}_X = \tfrac{1}{N}\sum_i \delta_{x^i}.$$

Moreover $\inf_v L_{\mathrm{unc}} > 0$ *whenever* $x^i \neq x^j$ *for some* $i \neq j$.

Corollary 1 (The correct conditional/unconditional distinction)**.** *Both exact population flows are supported on the training set:* $p_1^{\mathrm{cond}}(\cdot \mid y^i) = \delta_{x^i}$ *and* $p_1^{\mathrm{unc}} = \frac{1}{N}\sum_i \delta_{x^i}$. *The distinction is therefore not "memorises vs. does not" but* single-example *vs.* full-empirical-measure *memorisation, with observable difference in the conditional second moment:*

$$\mathrm{Cov}[p_1^{\mathrm{cond}}(\cdot \mid y^i)] = 0 \qquad \textit{vs.} \qquad \mathrm{Cov}[p_1^{\mathrm{unc}}] = \widehat{\Sigma}_X. \tag{2}$$

The often-cited observation $\mathrm{tr}\,\mathrm{Cov} \approx \mathrm{tr}\,\widehat{\Sigma}_X$ for the unconditional baseline is precisely (2), i.e. evidence of full-empirical-measure memorisation, *not* of non-memorisation.

3.2 Label smoothing is exactly kernel regression

The natural fix for a label that identifies its example too precisely is to blur it: replace y^i by a noisy $\widetilde{Y} = y^i + h\varepsilon$ and see what the exact minimiser does with the resulting ambiguity. We show this is not a heuristic smoothing trick but an exact statement: the population minimiser becomes the closed-form velocity field of a Nadaraya–Watson kernel regression over the training atoms.

Proposition 4 (Kernel-weighted population minimiser)**.** *For* $h > 0$, $t < 1$,

$$v_h^\star(x,t,y) = \sum_i w_i^{(h)}(x,t,y)\,\frac{x^i - x}{1-t}, \qquad w_i^{(h)} = \frac{\pi_0\big(\frac{x-tx^i}{1-t}\big) K_h(y-y^i)}{\sum_j \pi_0\big(\frac{x-tx^j}{1-t}\big) K_h(y-y^j)}. \tag{3}$$

Proof. Fix (x,t,y) with $t<1$. By Bayes' rule applied to the discrete latent index I,

$$\Pr(I = i \mid X_t = x, t, \widetilde{Y} = y) = \frac{\Pr(I=i)\, p(x,y \mid I=i,t)}{\sum_j \Pr(I=j)\, p(x,y \mid I=j,t)}. \tag{4}$$

Given $I=i$, $X_t = (1-t)X_0 + tx^i$ is a deterministic affine function of X_0 alone and $\widetilde{Y} = y^i + h\varepsilon$ is a deterministic affine function of ε alone, with $X_0 \perp \varepsilon$; hence $X_t \perp \widetilde{Y} \mid I = i$, with marginal conditional densities $p(x \mid I=i,t) = (1-t)^{-d}\pi_0\big(\frac{x-tx^i}{1-t}\big)$ (Lemma 4) and $p(y \mid I=i) = K_h(y-y^i)$ (the density of $y^i + h\varepsilon$). Independence gives the joint conditional density $p(x,y \mid I = i,t) = (1-t)^{-d}\pi_0\big(\frac{x-tx^i}{1-t}\big)K_h(y-y^i)$; with the uniform prior $\Pr(I=i) = 1/N$, (4) becomes

$$\Pr(I = i \mid X_t = x, t, \widetilde{Y} = y) = \frac{\frac{1}{N}(1-t)^{-d}\,\pi_0\big(\frac{x-tx^i}{1-t}\big)K_h(y-y^i)}{\sum_j \frac{1}{N}(1-t)^{-d}\,\pi_0\big(\frac{x-tx^j}{1-t}\big)K_h(y-y^j)} = \frac{\pi_0\big(\frac{x-tx^i}{1-t}\big)K_h(y-y^i)}{\sum_j \pi_0\big(\frac{x-tx^j}{1-t}\big)K_h(y-y^j)} = w_i^{(h)}(x,t,y),$$

the i-independent factor $\frac{1}{N}(1-t)^{-d}$ cancelling between the numerator and every term of the denominator (the kernel normalisation constant inside K_h cancels identically for the same reason). Given $I=i$, one further conditions on $X_t = x$: by Lemma 4 this determines $X_0 = (x - tx^i)/(1-t)$, so $U = X_1 - X_0 = x^i - X_0 = (x^i - x)/(1-t)$, exactly as in Proposition 2(a). By Proposition 1, the population minimiser is $v_h^\star(x,t,y) = \mathbb{E}[U \mid X_t = x, t, \widetilde{Y} = y]$, obtained by averaging this per-index target over the index posterior just computed:

$$v_h^\star(x,t,y) = \sum_{i=1}^N \Pr(I=i \mid X_t = x, t, \widetilde{Y} = y)\cdot\frac{x^i-x}{1-t} = \sum_{i=1}^N w_i^{(h)}(x,t,y)\,\frac{x^i - x}{1-t},$$

which is (3). □

Equation (3) factorises the index posterior into a spatial source-consistency term $\pi_0(\frac{x-tx^i}{1-t})$ and a label kernel $K_h(y-y^i)$. Write $p_i^{(h)}(y) := K_h(y-y^i)/\sum_j K_h(y-y^j)$ for the normalised label weights, i.e. the index posterior obtained from the label factor alone; the next lemma shows $v_h^\star$ is exactly the conditional velocity of an explicit, alternative coupling driven only by these weights, which is what lets us read off the endpoint law in Theorem 1 below.

Lemma 1 (Equivalent mixture coupling)**.** *Fix* y *and consider the coupling* $I \sim p^{(h)}(y)$, $X_0 \sim \pi_0$ *(independent of* I*),* $X_1 := x^I$, *with the same linear interpolant* $X_t = (1-t)X_0 + tX_1$. *Its conditional velocity field* $\mathbb{E}[X_1 - X_0 \mid X_t = x, t]$ *equals* $v_h^\star(x,t,y)$ *of* (3) *for every* $x, t < 1$.

Proof. Under this coupling, $\Pr(I = i \mid X_t = x, t) \propto \Pr(I = i)\, p(x \mid I = i, t) = p_i^{(h)}(y) \cdot (1 - t)^{-d}\pi_0\big(\frac{x - tx^i}{1-t}\big)$ by the same Bayes'-rule computation as above (now with prior $p_i^{(h)}(y)$ in place of $1/N$), and $p_i^{(h)}(y) \propto K_h(y - y^i)$ by definition, so $\Pr(I = i \mid X_t = x, t) \propto \pi_0\big(\frac{x-tx^i}{1-t}\big) K_h(y - y^i)$—the same proportionality (in i, at fixed x, t, y) as the numerator of $w_i^{(h)}$ in (3); normalising over i recovers $w_i^{(h)}(x, t, y)$ exactly. As in the proof above, given $I = i$ the target is $(x^i - x)/(1 - t)$, so $\mathbb{E}[X_1 - X_0 \mid X_t = x, t] = \sum_i w_i^{(h)}(x, t, y)(x^i - x)/(1 - t) = v_h^\star(x, t, y)$. □

Theorem 1 (Endpoint law under label smoothing)**.** *For every $h > 0$ and query y,*

$$p_h^{\text{gen}}(\cdot \mid y) = \sum_{i=1}^{N} p_i^{(h)}(y)\, \delta_{x^i}. \tag{5}$$

Proof. By Lemma 1, $v_h^\star(\cdot, \cdot, y)$ is exactly the conditional velocity field of the mixture coupling $I \sim p^{(h)}(y)$, $X_0 \sim \pi_0$, $X_1 = x^I$, driven by the same linear interpolant. This field has the form $v(x, t) = \sum_i w_i(x, t)(x^i - x)/(1 - t)$ of Lemma 5, with smooth weights $w_i = w_i^{(h)}(x, t, y)$ summing to one (a ratio of smooth, strictly positive functions of (x, t), by Lemma 4's positivity of π_0), so Lemma 5 applies: the ODE $\dot{x}_t = v_h^\star(x_t, t, y)$ is well posed on $[0, 1)$ for every initial condition, and by the superposition principle its flow $(\Phi_t)_{t<1}$ transports $\pi_0 = \mathrm{Law}(X_0)$ to $p_t := \mathrm{Law}(X_t)$ for every $t < 1$, where X_t is the coupling's own interpolant—because $v_h^\star$ is by construction the coupling's conditional velocity, $(\Phi_t)_\# \pi_0$ and $\mathrm{Law}(X_t)$ solve the same continuity equation with the same initial condition, hence coincide by uniqueness. It remains to identify the limiting law of X_t as $t \uparrow 1$: since $X_t = (1 - t)X_0 + tx^I$ with $X_0 \sim \pi_0$ independent of I, $X_t \to x^I$ almost surely as $t \uparrow 1$ for every realisation of I; hence $\mathrm{Law}(X_t) \Rightarrow \mathrm{Law}(x^I) = \sum_i p_i^{(h)}(y)\delta_{x^i}$ weakly, which is $p_1 = \lim_{t\uparrow 1} p_t$ by Lemma 5. This is (5). □

Corollary 2 (Zero bandwidth at fixed N)**.** *If y has a unique nearest label $i^\star = \arg\min_i |y - y^i|$, then $p_i^{(h)}(y) \to \mathbf{1}\{i = i^\star\}$ as $h \to 0$, so $p_h^{\text{gen}}(\cdot \mid y) \Rightarrow \delta_{x^{i^\star}}$; at $y = y^i$ this recovers Proposition 2.*

Proof. For $i \neq i^\star$, $K_h(y - y^i)/K_h(y - y^{i^\star}) = \exp\big[-(|y - y^i|^2 - |y - y^{i^\star}|^2)/2h^2\big] \to 0$ as $h \to 0$, since $|y - y^i| > |y - y^{i^\star}|$; normalising the weights $K_h(y - y^j)$ gives $p^{(h)}(y) \to \mathbf{1}\{i = i^\star\}$, and Theorem 1 gives the claim. □

Corollary 3 (Infinite bandwidth at fixed N)**.** *$p_i^{(h)}(y) \to 1/N$ for every i as $h \to \infty$, so $p_h^{\text{gen}}(\cdot \mid y) \Rightarrow p_1^{\text{unc}}$.*

Proof. $K_h(y - y^i) = (2\pi h^2)^{-k/2}\big[1 + O(h^{-2})\big]$ uniformly in i as $h \to \infty$, so the normalised weights $p_i^{(h)}(y) \to 1/N$; apply Theorem 1. □

Thus h interpolates continuously between the two hard regimes of Section 3.1:

$$\delta_{x^{i^\star}} \xrightarrow{\;h\nearrow\;} \sum_i p_i^{(h)}(y)\, \delta_{x^i} \xrightarrow{\;h\to\infty\;} \tfrac{1}{N}\sum_i \delta_{x^i}.$$

Proposition 5 (Exact finite-N moments)**.** $\bar{x}_h(y) = \sum_i p_i^{(h)}(y)\, x^i$ *and* $\mathrm{Cov}_h(y) = \sum_i p_i^{(h)}(y)\,(x^i - \bar{x}_h(y))(x^i - \bar{x}_h(y))^\top$*, both exactly computable from the training set for any h. They are the correct reference against which generated moments must be compared.* $\mathrm{tr}\,\mathrm{Cov}_h(y)$ *need not be monotone in h; monotonicity is not implied and must be reported empirically.*

Proposition 6 (Label smoothing never leaves the training set)**.** *For every $h \in [0, \infty)$,* $\mathrm{supp}(p_h^{\text{gen}}(\cdot \mid y)) \subseteq \{x^1, \ldots, x^N\}$*. Consequently, if the true posterior $p(\cdot \mid y)$ is absolutely continuous,*

$$W_2^2\big(p_h^{\text{gen}}(\cdot \mid y),\, p(\cdot \mid y)\big) \;\geq\; \int \mathrm{dist}\left(x, \{x^i\}\right)^2 p(x \mid y)\, dx \;>\; 0, \tag{6}$$

and this lower bound does not depend on h.

Proof. The support claim is Theorem 1. For any coupling (X, X') of p_h^{gen} and $p(\cdot \mid y)$, $X \in \{x^i\}$ a.s. so $|X - X'| \geq \operatorname{dist}(X', \{x^i\})$ pointwise; take expectations and infimise. The right side is the same for every h and is strictly positive because a finite set is $p(\cdot \mid y)$-null. □

Proposition 6 is the central caveat on the "variance is restored at $h \approx 0.1$" claim: label smoothing *redistributes weight over training atoms*; it never generates new samples. Matching $\operatorname{tr}\operatorname{Cov}_h$ matches a second moment, not the posterior. A genuine remedy requires $h \to 0$ *and* $N \to \infty$ jointly, which is the Nadaraya–Watson consistency regime of Proposition 8 below—not h alone.

Proposition 7 (Bandwidth expansion of the conditional covariance)**.** *Let (x^j, y^j) be i.i.d. with $\Sigma(y) = \operatorname{Cov}(X \mid Y = y)$ and $J(y) = \partial\mu/\partial y$. Under standard smoothness and $Nh^k \to \infty, h \to 0$,*

$$\operatorname{Cov}_h(y) = \Sigma(y) + h^2 J(y)J(y)^\top + O(h^4) + O_P\big((Nh^k)^{-1/2}\big), \tag{7}$$

so $\operatorname{tr}\operatorname{Cov}_h(y) = \operatorname{tr}\Sigma(y) + h^2\|J(y)\|_F^2 + \cdots$.

The between-group inflation $h^2\|J\|_F^2$ is positive and, unlike the bias of the conditional *mean*, is not annihilated by kernel symmetry; it produces the right-hand (over-smoothing) branch of the empirical U-curve. In the linear-Gaussian model $\mu(y)$ is exactly linear, so $J = \Sigma_{\text{post}}A^\top/\sigma_{\text{obs}}^2$ is known in closed form and the $O(h^4)$ term vanishes: (7) is then a parameter-free prediction.

Proposition 8 (Nadaraya–Watson rates for the conditional mean)**.** *Under the assumptions of Proposition 7, $\bar{x}_h(y)$ is exactly the Nadaraya–Watson estimator of $\mu(y)$, with*

$$\text{Bias} = O(h^2), \qquad \text{Var} = O\big((Nh^k)^{-1}\big), \qquad \text{MSE} = O(h^4) + O\big((Nh^k)^{-1}\big),$$

minimised at $h^\star \asymp N^{-1/(k+4)}$.

Proof. The bias is the $O(h^2)$ term identified in the proof of Proposition 7 (Taylor-expanding $\mu(y^j)$ about y against the symmetric kernel weights); squaring gives $\text{Bias}^2 = O(h^4)$. Averaging over an effective sample size $\asymp Nh^k$ gives the usual Nadaraya–Watson variance rate $O((Nh^k)^{-1})$. Summing, $\text{MSE}(h) = c_1h^4 + c_2(Nh^k)^{-1} + o(\cdot)$ for constants $c_1, c_2 > 0$; differentiating and setting to 0, $4c_1h^3 = c_2kN^{-1}h^{-k-1}$, i.e. $h^{k+4} \propto N^{-1}$, so $h^\star \asymp N^{-1/(k+4)}$. □

Queries at $y = y^i$ are *design points*: x^i receives the maximal weight $K_h(0)$, and $Nh^k \to \infty$ is exactly what makes its relative influence vanish; without it Proposition 8 does not apply there. This is a genuinely different $h \to 0$ limit from Corollary 2's fixed-N nearest-neighbour collapse: same symbol, opposite conclusions.

3.3 Stochastic interpolant noise does not help

Label noise is not the only place one could inject randomness to try to break the hard identification of Section 3.1: an equally natural candidate is to add noise to the interpolant path itself, as in stochastic-interpolant constructions. We show this alternative remedy is qualitatively different: it never touches the population endpoint, at any noise level.

Replace the interpolant by $X_t = (1-t)X_0 + tX_1 + \gamma(t)Z$ with $Z \sim \mathcal{N}(0, I_d)$ independent and $\gamma(t) = \sigma\sqrt{t(1-t)}$, so $\gamma(0) = \gamma(1) = 0$. A single Z defines the whole trajectory, and the correct pathwise-derivative target is

$$\dot{X}_t = X_1 - X_0 + \dot{\gamma}(t)Z, \qquad \dot{\gamma}(t) = \frac{\sigma(1-2t)}{2\sqrt{t(1-t)}}. \tag{8}$$

Regressing on $X_1 - X_0$ alone is a *different* objective and does not yield the marginal velocity of the noisy path. Let $s_t^2 := (1-t)^2 + \gamma(t)^2 = (1-t)[(1-t) + \sigma^2 t]$, so $s_0 = 1,\ s_1 = 0$.

Proposition 9 (Closed form and endpoint invariance in σ)**.** *Let $h = 0$ with distinct labels and $v_\sigma^\star = \mathbb{E}[\dot{X}_t \mid X_t, t, Y]$ for target (8). Then for $t < 1$: (a) $v_\sigma^\star(x, t, y^i) = x^i + c(t)(x - tx^i)$ with $c(t) = \frac{-(1-t) + \frac{\sigma^2}{2}(1-2t)}{s_t^2} = \frac{1}{2}\frac{d}{dt}\log s_t^2$; (b) the flow is $x_t = t\,x^i + s_t\,x_0$; and (c) since $s_1 = 0$,*

$$p_1^{\text{cond}}(\cdot \mid y^i) = \delta_{x^i} \quad \textit{for every } \sigma \geq 0. \tag{9}$$

Proof. Given $I = i$, write $r := x - tx^i$. Since $X_t = (1-t)X_0 + tx^i + \gamma(t)Z$, $r = (1-t)X_0 + \gamma(t)Z$, a linear combination of the independent standard Gaussians $X_0, Z \sim \mathcal{N}(0, I_d)$, so $r \sim \mathcal{N}(0, s_t^2 I_d)$ with $s_t^2 = (1-t)^2 + \gamma(t)^2$. For jointly Gaussian (X_0, r), $\mathrm{Cov}(X_0, r) = (1-t)I_d$ (the $\gamma(t)Z$ term is independent of X_0), so the standard Gaussian conditional-mean formula gives $\mathbb{E}[X_0 \mid r] = \mathrm{Cov}(X_0, r)\ \mathrm{Var}(r)^{-1} r = \frac{1-t}{s_t^2} r$; identically, $\mathrm{Cov}(Z, r) = \gamma(t) I_d$ gives $\mathbb{E}[Z \mid r] = \frac{\gamma(t)}{s_t^2} r$. Hence

$$v_\sigma^\star(x, t, y^i) = \mathbb{E}[\dot{X}_t \mid X_t = x, t, I = i] = x^i - \mathbb{E}[X_0 \mid r] + \dot\gamma(t)\,\mathbb{E}[Z \mid r] = x^i + \frac{-(1-t) + \dot\gamma(t)\gamma(t)}{s_t^2}\, r.$$

Since $\gamma(t)^2 = \sigma^2 t(1-t)$, differentiating gives $2\dot\gamma\gamma = \sigma^2(1-2t)$, i.e. $\dot\gamma\gamma = \frac{\sigma^2}{2}(1-2t)$, which yields (a)'s expression for $c(t)$. For the logarithmic-derivative identity, $\frac{d}{dt} s_t^2 = -2(1-t) + 2\dot\gamma\gamma = 2 s_t^2 c(t)$ by the same computation, so $c(t) = \frac{1}{2}\frac{d}{dt}\log s_t^2$.

(b) Along the flow, $u_t := x_t - tx^i$ satisfies $\dot u_t = \dot x_t - x^i = v_\sigma^\star(x_t, t, y^i) - x^i = c(t) u_t$ by (a) (with $r = u_t$ evaluated on the trajectory), a linear ODE with solution $u_t = u_0 \exp\big(\int_0^t c(\tau)\, d\tau\big) = x_0 \exp\big(\frac{1}{2}[\log s_t^2 - \log s_0^2]\big) = x_0 s_t$, using $s_0 = 1$ and $u_0 = x_0 - 0 \cdot x^i = x_0$. Hence $x_t = tx^i + s_t x_0$.

(c) Since $s_1 = (1-1)[(1-1) + \sigma^2 \cdot 1] = 0$, (b) gives $x_1 = x^i$ for every $x_0 \sim \pi_0$ and every $\sigma \geq 0$, so $p_1^{\mathrm{cond}}(\cdot \mid y^i) = \delta_{x^i}$. ∎

Corollary 4 (What interpolant noise does and does not change)**.** *Interpolant noise does not change the population endpoint law* (9) *for any σ; it* does *raise the irreducible error* ($\inf_v L_\sigma > 0$ *for* $\sigma > 0$, *whereas* $\inf_v L_0 = 0$) *and halves the Lipschitz blow-up rate:* $\lim_{t\uparrow 1}(1-t)|c(t)| = 1$ *for* $\sigma = 0$ *and* $\frac{1}{2}$ *for* $\sigma > 0$.

Proof. Endpoint invariance is Proposition 9(c).

Irreducible error. For $\sigma = 0$, $\gamma \equiv 0$ so $\dot X_t = X_1 - X_0$ is exactly the target of Proposition 2, giving $\inf_v L_0 = 0$. For $\sigma > 0$: by the same projection argument as Proposition 1, applied to $\dot X_t$ and (X_t, t, Y), it suffices to show $\mathrm{Var}(\dot X_t \mid X_t = x, t, I = i) > 0$ for some positive-measure set of t. Writing $\dot X_t - x^i = -X_0 + \dot\gamma(t) Z$ and $r = (1-t)X_0 + \gamma(t)Z$ as in the proof of Proposition 9, the standard Gaussian conditioning formula gives, for $u = (-1, \dot\gamma(t))$ and $w = (1-t, \gamma(t))$,

$$\mathrm{Var}(\dot X_t \mid r, I = i) = |u|^2 - \frac{(u \cdot w)^2}{|w|^2} = \big(1 + \dot\gamma(t)^2\big) - c(t)^2 s_t^2,$$

using $u \cdot w = -(1-t) + \dot\gamma(t)\gamma(t) = c(t) s_t^2$ from the proof of Proposition 9. This vanishes only if $u \parallel w$, i.e. $\dot\gamma(t)(1-t) = -\gamma(t)$; substituting $\gamma(t) = \sigma\sqrt{t(1-t)}$ and $\dot\gamma(t) = \sigma(1-2t)/(2\sqrt{t(1-t)})$ reduces this to $1 - 2t = -2t$, i.e. $1 = 0$—impossible. So for $\sigma > 0$ the residual variance is strictly positive for *every* $t \in (0,1)$, and $\inf_v L_\sigma(v) = \mathbb{E}[\mathrm{Var}(\dot X_t \mid X_t, t, Y)] > 0$.

Lipschitz rate. From (a), $(1-t)c(t) = \dfrac{-(1-t) + \frac{\sigma^2}{2}(1-2t)}{(1-t) + \sigma^2 t}$, which as $t \uparrow 1$ tends to -1 if $\sigma = 0$ and to $-\sigma^2/2 \,/\, \sigma^2 = -\frac{1}{2}$ if $\sigma > 0$, giving the stated limits. ∎

Proposition 10 (The two mechanisms act on different factors)**.** *In the factorisation* (3)*, label smoothing modifies the label factor $K_h(y - y^i)$ and therefore changes p_1 (Theorem 1); interpolant noise rescales the spatial factor* $\mathcal{N}(tx^i, s_t^2 I_d)$ isotropically and identically for every i, *leaving the endpoint law unchanged (Proposition 9). The two remedies are therefore not interchangeable.*

Proof. By Theorem 1, for $h > 0$ (and $\sigma = 0$) the endpoint law $p_h^{\mathrm{gen}}(\cdot \mid y) = \sum_i p_i^{(h)}(y)\delta_{x^i}$ depends on h only through the label factor $K_h(y - y^i)$ in (3), and Corollaries 2–3 show it genuinely interpolates between two distinct endpoint laws as h ranges over $[0, \infty)$. By contrast, for $h = 0$ with distinct labels and any $\sigma \geq 0$, $Y = y^i$ already pins $I = i$ deterministically, so no index-posterior argument is needed: Proposition 9 shows directly that although the spatial factor of Lemma 4, $\mathcal{N}(tx^i, (1-t)^2 I_d)$, is replaced by $\mathcal{N}(tx^i, s_t^2 I_d)$ for *every* i identically, the endpoint is exactly invariant, $p_1^{\mathrm{cond}}(\cdot \mid y^i) = \delta_{x^i}$ for every σ. Since perturbing the label factor changes p_1 while perturbing the spatial factor identically across i does not, the two mechanisms act through different, non-interchangeable channels. ∎

This cleanly separates the present conditional mechanism from the unconditional "resample-x_0" remedy of the gradient-variance line (Reu et al., 2025), which operates through attainability rather than through the population optimum.

3.4 Finite capacity: a genuine representation floor

The results so far describe the population minimiser over *all* square-integrable fields; a trained network is restricted to a specific, finite-capacity class. We now give one concrete way this restriction alone forces a positive loss floor, independent of any optimisation difficulty.

Proposition 11 (Lipschitz lower bound)**.** *Fix i, $t < 1$, $f_i(x,t) = (x^i - x)/(1-t)$, and suppose $v(\cdot, t, y^i)$ is L-Lipschitz in x. Under $X_t \mid I = i \sim \mathcal{N}(tx^i, (1-t)^2 I_d)$, $\mathbb{E}[|v(X_t, t, y^i) - f_i(X_t, t)|^2 \mid I = i] \geq d\,[1 - L(1-t)]_+^2$.*

Corollary 5 (Loss floor under a uniform Lipschitz constraint)**.** *For $\mathcal{F}_L = \{v : \mathrm{Lip}_x\, v(\cdot, t, y^i) \leq L\ \forall t, i\}$ with $L \geq 1$ and $t \sim \mathrm{Unif}(0,1)$,*

$$\inf_{v \in \mathcal{F}_L} L_0(v) \;\geq\; d \int_{1-1/L}^{1} [1 - L(1-t)]^2\, dt \;=\; \frac{d}{3L} \;>\; 0. \tag{10}$$

Two warnings temper (10). First, $L^\star_{\text{model}} > 0$ requires the constraint: an MLP with unbounded weights lies in no fixed $\mathcal{F}_L$, and by universal approximation $\inf_\theta L(v_\theta)$ may be 0. The correct statement is conditional: any architecture with $\mathrm{Lip}_x\, v_\theta \leq L$ cannot beat $d/(3L)$. Second, in our experiments ($d = 2$, plateau $L_{\text{trained}} \approx 0.48$ at the 2×10^5-iteration checkpoint used for the Lipschitz measurement below) the bound demands only $L \geq 1.39$, whereas a trained network has Lip_x of order 10^1–10^2, giving a floor of order 10^{-2}—one to two orders of magnitude below the plateau. Corollary 5 is thus best used as a *measurement*: estimate L, compare $d/(3L)$ to L_{trained}, and report which gap dominates (Section 4.5).

3.5 Finite-sample memorisation and exact non-representability

This subsection translates two representability results from the unconditional gradient-variance/stochastic-interpolant literature (Reu et al., 2025) into the present notation: (i) why a fixed finite *unconditional* batch already admits a zero-loss memorising field despite Proposition 3's positive population loss, and (ii) why no finite network can exactly represent the collapsed field ($\star$). Unlike Sections 3.1–3.4, the finite sample itself is treated as random here.

Lemma 2 (Generic non-intersection of finite interpolant segments)**.** *Let $x_0^1, \ldots, x_0^N \overset{\text{iid}}{\sim} \pi_0$ and, independently, $z^1, \ldots, z^N \overset{\text{iid}}{\sim} \rho_1$ be N unconditional source–target pairs (no labels), both laws absolutely continuous on $\mathbb{R}^d$, and $\ell_i(t) = (1-t)x_0^i + tz^i$, $t \in [0,1]$. (a) If $d \geq 2$: for any fixed $i \neq j$, $\Pr[\exists\, t \in (0,1) : \ell_i(t) = \ell_j(t)] = 0$. (b) If $d > 2$: for any fixed $i \neq j$, $\Pr[\exists\, t_i, t_j \in (0,1),\, t_i \neq t_j : \ell_i(t_i) = \ell_j(t_j)] = 0$.*

Proof. (a) Fix $\hat{t} \in (0,1)$; $\ell_i(\hat{t}) = \ell_j(\hat{t})$ solves to $z^j = \frac{1-\hat{t}}{\hat{t}}(x_0^i - x_0^j) + z^i$, which pins z^j to a one-dimensional affine subspace of $\mathbb{R}^d$ determined by $(x_0^i, x_0^j, z^i, \hat{t})$. This subspace is Lebesgue-null whenever $d \geq 2$, and ρ_1 is absolutely continuous, so conditionally on (x_0^i, x_0^j, z^i) the event has probability 0; integrating over (x_0^i, x_0^j, z^i) preserves probability 0, and the bound is uniform in $\hat{t}$, so the same holds for "some $t = \hat{t}$" (the constraint is on z^j alone and does not depend on which $\hat{t}$ occurs). A union bound over the finitely many pairs (i,j) finishes (a). (b) With $t_i \neq t_j$ both free, solving for z^j pins it to a two-dimensional affine subspace, Lebesgue-null when $d > 2$; the rest of the argument is identical. $\square$

Proposition 12 (A memorising field always exists on a fixed finite unconditional batch)**.** *Fix N unconditional pairs $\{(x_0^i, z^i)\}_{i=1}^N$ as in Lemma 2 with $d > 2$ and the z^i pairwise distinct a.s., and sample m time points $t^{(i,j)} \in (0,1)$, $j = 1, \ldots, m$, independently across (i,j). Let $X_t^{(i,j)} = (1 - t^{(i,j)})x_0^i + t^{(i,j)} z^i$ and*

$$L_{\text{MC}}^{\text{unc}}(v) = \frac{1}{Nm} \sum_{i=1}^{N} \sum_{j=1}^{m} \left|(z^i - x_0^i) - v\big(X_t^{(i,j)}, t^{(i,j)}\big)\right|^2.$$

Then, almost surely over the draw of $\{(x_0^i, z^i)\}$ and $\{t^{(i,j)}\}$, some v attains $L_{\mathrm{MC}}^{\mathrm{unc}}(v) = 0$—even though $\inf_v L_{\mathrm{unc}}(v) > 0$ at the population level whenever the z^i are not all equal (Proposition 3).

Proof. By Lemma 2(a)–(b), union-bounded over the $\binom{N}{2}$ pairs, almost surely no two of the N full segments $\{\ell_i(t) : t \in (0,1)\}$ intersect at all, so in particular the Nm sampled points $(X_t^{(i,j)}, t^{(i,j)})$ never collide across different i. On this full-measure event $(z,t) \mapsto i$ is well defined on the sample, so setting $v(z,t) := z^{i(z,t)} - x_0^{i(z,t)}$ on the sample and $v := 0$ elsewhere gives $v(X_t^{(i,j)}, t^{(i,j)}) = z^i - x_0^i$ exactly for every sampled (i,j), so every summand of $L_{\mathrm{MC}}^{\mathrm{unc}}$ vanishes. □

Remark 1 (Why SGD does not land here). Proposition 12 is valid only for one *fixed* finite sample. Standard CFM training resamples x_0^i (and the time samples) independently every epoch, so the lookup table fit at epoch k is checked, at epoch $k+1$, against points it never saw; no single lookup table stays consistent with a growing stream of (x_0, t) draws. This is the precise mechanism behind the informal claim that resampling x_0—not mere continuity of π_0—is what pushes unconditional training toward the population optimum of Proposition 3 rather than a per-example memorising solution, and it gives a concrete reading of P6 (Table 3): at fixed capacity, larger N makes the epoch-consistent lookup table harder to realise with a smooth finite-parameter function, pushing the attained solution back toward the population optimum. Note the asymmetry with Proposition 2: conditional collapse needs no genericity argument and no fixed-batch caveat, because y^i pins down i deterministically for every x_0, not merely with probability 1 on one finite draw—resampling x_0 every epoch does nothing to prevent it, which is exactly why conditional collapse is universal while unconditional memorisation is not.

Lemma 3 (The collapse field cannot be exactly represented by a finite network)**.** *Let $g(t) = 1/(1-t)$ on $[0,1)$ and let $f_\theta : \mathbb{R} \to \mathbb{R}$ be any finite-depth, finite-width feedforward network with finite real weights, built from affine layers and globally Lipschitz activations (this covers ReLU, leaky-ReLU, tanh, sigmoid, GELU/SiLU, softplus, and every activation used in our experiments). Then $\sup_{t\in[0,1)} |f_\theta(t) - g(t)| = \infty$.*

Proof. A finite composition of affine maps (finite weight matrices, hence finite operator norm) and globally Lipschitz activations is globally Lipschitz on $\mathbb{R}$, with constant equal to the product of the finitely many layer constants. A Lipschitz function is bounded on any bounded set: fixing $t_0 \in [0,1)$, $|f_\theta(t)| \le |f_\theta(t_0)| + K|t - t_0| \le |f_\theta(t_0)| + K$ for all $t \in [0,1)$, so $\sup_{t\in[0,1)} |f_\theta(t)| < \infty$. But $g(t) \to \infty$ as $t \uparrow 1$, so g is unbounded on $[0,1)$, hence so is $f_\theta - g$. □

For $x \ne x^i$ fixed, $t \mapsto v_0^\star(x,t,y^i) = (x^i - x)/(1-t)$ has exactly this $1/(1-t)$ blow-up along the direction $x^i - x$, so Lemma 3 (applied coordinatewise to that direction) shows that *no* finite network of the kind used in Section 4 can represent $v_0^\star$ *exactly* on all of $[0,1)$: for every fixed parameter vector θ there is some (x,t) at which $v_\theta \ne v_0^\star$. This does not contradict universal approximation, which guarantees only *uniform* approximation on *compact* sets: every $[0, 1-\delta]$, $\delta > 0$, is disjoint from the singularity at $t = 1$—exactly the regime the sampler must traverse to reach the collapse point x^i, and exactly where the ODE solver's $t = 1$ handling matters in practice. We stop short, however, of claiming this pointwise non-representability forces $L_{\mathrm{model}}^\star - L_{\mathrm{all}}^\star > 0$ for an architecture whose weights (hence Lipschitz constant) are unconstrained: pointwise blow-up on a shrinking neighbourhood of $t = 1$ is compatible, in principle, with a sequence of increasingly ill-conditioned finite networks driving the *expected* loss $L_0(v_\theta)$ arbitrarily close to 0 (consistent with the growing empirical Lipschitz constant we measure in Figure 12). The genuine, quantitative loss floor we prove, $d/(3L)$, therefore applies only to the Lipschitz-capped class $\mathcal{F}_L$ of Corollary 5, and should not be read as a statement about arbitrary finite architectures.

3.6 Scope: what is and is not predicted

The population theory proves exact consequences for the empirical minimiser (Propositions 2–8, Corollaries 2 and 3, Proposition 11, Corollary 5). Section 3.5 additionally proves exact statements about a *randomly drawn* finite unconditional batch (Lemma 2, Proposition 12, Lemma 3)—these are not part of the fixed-dataset empirical-population characterisation above, but are exact nonetheless. It does *not* determine: the number of SGD steps to approach the optimum; the dependence of *observed* collapse on N at fixed architecture (the population optimum collapses for every finite N, so a non-collapsing N=5000 run probes representation/optimisation, it does not contradict Proposition 2); the dependence of observed collapse on σ_{obs} (the theory implies no monotonicity, so a flat sweep is *consistent with*, not a falsification of, the theory); the monotonicity of $\operatorname{tr}\operatorname{Cov}_h$ in h; or finite-sample error bars.

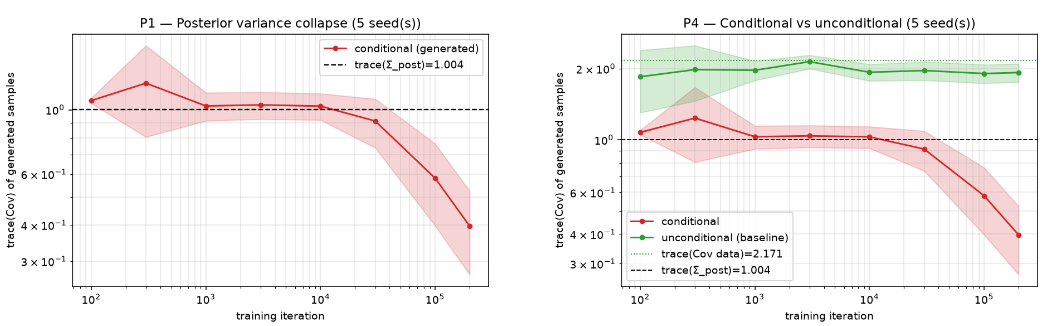
> *Phrasing rule. A quantity in the second list is never a "theoretical prediction" that "matched" or "failed." Correct form: the exact population theory does not determine X; the observed behaviour of X is an empirical finding. We follow this rule in Section 4, marking such results "(empirical)."*

4 Experiments

Setup. Unless noted, EXP-1 uses the linear-Gaussian problem d=2, k=1, N=200, σ_{obs}=0.1, prior and source $\mathcal{N}(0, I)$, deterministic linear interpolant (σ=0); a 4-layer width-128 MLP with SiLU and a sinusoidal time embedding (dim 64); Adam at lr=10^{-3}, batch 256; RK4 integration with 100 steps stopping at t=$1-10^{-3}$; evaluation over M=1000 samples at 20 training conditions and 8 held-out conditions. We do *not* early-stop: overtraining is the object of study. Sanity checks pass first (analytic posterior satisfies the normal equations to 1.8×10^{-15}; closed-form field integration maps every $x_0 \to x^i$; the unconditional model does not leak y), so the collapse below is not an artefact of a y-leak or a broken ODE. Parameter sweeps use five seeds per point (seeds 0–2 on CPU, seeds 3–4 on 2×T4 GPU via the same training code, `device: auto`). The seed controls *both* the random draw of the problem instance (the forward operator A, hence Σ_{post}) *and* all training/data randomness, so the reported seed-to-seed spread reflects both sources jointly rather than SGD noise at a fixed problem alone; we did not find it necessary to disentangle the two for the conclusions drawn here, but flag this as a conflation a reader should be aware of when interpreting the error bars below.

4.1 EXP-1: Exact-theory quantities (P1–P4)

Figure 1 summarises the four exact-theory observables. As training proceeds past the good-calibration phase, the conditional generated variance collapses from $\approx \Sigma_{\text{post}}$ toward 0 (P1), the velocity field approaches the collapsed closed form $(x^i - x)/(1-t)$ (P2), the generated mean moves from μ_{post} toward the training point x^i (P3), and the unconditional baseline retains data-scale variance throughout (P4).



(a) P1: conditional variance collapse.

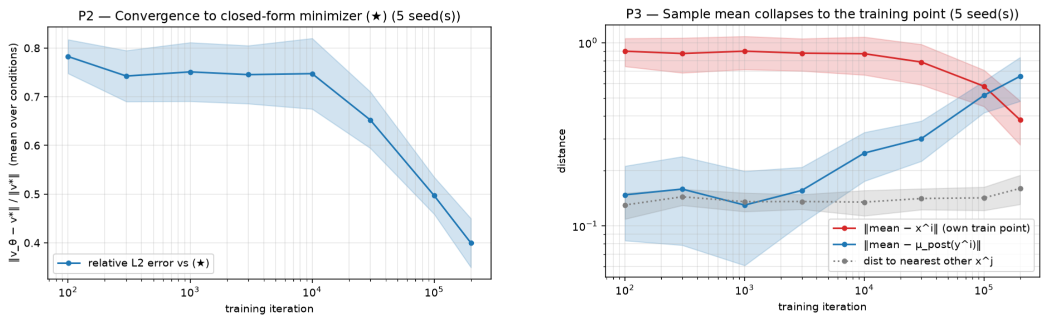
(b) P4: conditional vs. unconditional.

Figure 1: Overtraining collapses the conditional generated variance toward zero, while the unconditional model retains data-scale variance (Corollary 1).

The variance tracks the true posterior up to $\sim 10^4$ iterations—the model is a correct Bayesian sampler in this phase, which is exactly why early stopping works (Dasgupta et al., 2026)—and then collapses. At 2×10^5 the variance has not yet reached 0 but stalls near 0.40. A single-seed extended run at a *fixed* learning rate shows all four observables decreasing together with the loss (to 0.16/0.36 at 7×10^5), so the residual at 2×10^5 is an *optimisation* gap, not a failure of Proposition 2—but that fixed-lr run eventually diverges near 10^6 iterations, since the target ($\star$) steepens as $t \to 1$ and a constant step size becomes too large once the loss is already small. Adding a cosine learning-rate

quantity (t=200k unless noted)	iteration 10^2	10^4	10^5	2×10^5	1×10^6
conditional $\operatorname{tr}\mathrm{Cov}$	1.08 ± 0.02	1.03 ± 0.11	0.58 ± 0.18	0.396 ± 0.127	0.120 ± 0.054
velocity rel. error vs. ($\star$)	0.78 ± 0.03	0.75 ± 0.07	0.50 ± 0.04	0.40 ± 0.05	0.168 ± 0.052
$\|\mathrm{mean} - x^i\|$	0.90 ± 0.16	0.87 ± 0.20	0.58 ± 0.13	0.38 ± 0.10	0.137 ± 0.065
$\|\mathrm{mean} - \mu_{\mathrm{post}}\|$	0.15 ± 0.06	0.25 ± 0.07	0.52 ± 0.10	0.66 ± 0.18	0.816 ± 0.172
unconditional $\operatorname{tr}\mathrm{Cov}$	1.85 ± 0.55	1.93 ± 0.16	1.91 ± 0.17	1.926 ± 0.171	—

Table 1: EXP-1, mean$\pm$std over 5 seeds, regenerated directly from the tracked metrics; the 1×10^6 column is the extended run *with* the cosine learning-rate schedule of Section 4.1 (5 seeds; no unconditional counterpart). Reference values: $\operatorname{tr}\Sigma_{\mathrm{post}} \approx 1.004$, $\operatorname{tr}\widehat{\Sigma}_X = 2.171$ (seed 0). All four conditional quantities descend together toward the δ_{x^i} limit throughout, with no sign of the divergence the fixed-lr single-seed run exhibits at this iteration count (Figure 4); the between-seed spread at 1×10^6 is large relative to the improvement over the 2×10^5 column, so this last column is best read as confirming stability rather than precisely quantifying additional depth (see Section 4.1).



(a) P2: velocity $\to$ ($\star$).

(b) P3: mean $\to x^i$, away from μ_{post}.

Figure 2: The learned velocity approaches the collapsed closed form ($\star$) (a) and the generated mean migrates from the Bayes posterior mean to the memorised training point (b). Both descend in lockstep with P1.

schedule (lr : $10^{-3} \to 10^{-6}$ over the run) removes this divergence entirely. The cleanest evidence is a like-for-like, same-seed comparison: at the identical 7×10^5-iteration checkpoint, seed 0 reaches $\operatorname{tr}\mathrm{Cov} = 0.128$ with the schedule versus 0.162 at a fixed learning rate, and continues to 0.115 by $1{\times}10^6$ with no sign of the fixed-lr run's eventual blow-up. Averaged over *5 seeds* (rerun with the schedule), the mean trajectory decreases from $\gtrsim 1$ to 0.120 ± 0.054 at $1{\times}10^6$ iterations; we report this mean mainly to show that no seed diverges, not as a tight improvement over the old single-seed value of 0.16—the between-seed standard deviation is large enough (individual seeds at $1{\times}10^6$ range from 0.035 to 0.186) that 0.16 is not clearly excluded by 0.120 ± 0.054, and this spread is

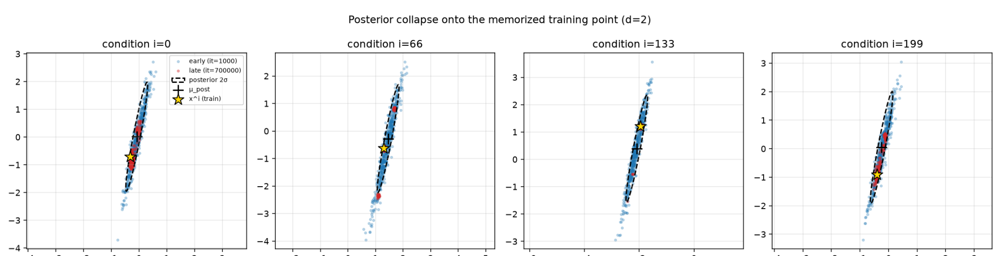


Figure 3: Qualitative d=2 collapse (seed 0, $7{\times}10^5$): the sample cloud that initially fills the posterior ellipse (blue, early) contracts to the memorised training point (red, late) for four different conditions y^i.

inflated by the seed-conflated problem instance noted in Section 4 (a harder or easier random A, not only training noise). The improvement is also not uniform across the last third of training: from 7×10^5 to 1×10^6—by which point the learning rate has decayed to $\lesssim 2 \times 10^{-4}$—four of five seeds improve and one worsens slightly, so this tail mostly protects against divergence rather than reliably deepening the collapse further. With that said, the memorisation ratio of the paragraph below reaches 0.955 ± 0.020 at 1×10^6—95% of generated samples individually satisfy the nearest-neighbour memorisation criterion, a low-variance and unambiguous confirmation that collapse continues. Taken together, this resolves the open question of Section 5: reaching deep collapse cleanly, without divergence, needs only a decaying learning rate, not a change to the theory, even though we cannot yet say precisely how much deeper a better-tuned schedule could push $\operatorname{tr}\operatorname{Cov}$.

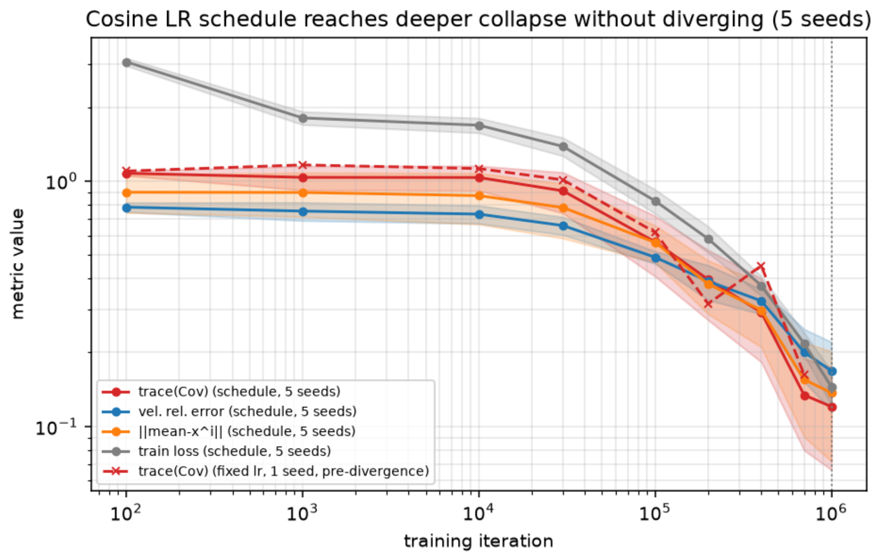


Figure 4: Extended training with a cosine learning-rate schedule (5 seeds, error bars over seeds) tracks all the way to 10^6 iterations without ever diverging, unlike the fixed-lr single seed (dashed), which is cut off at its last pre-divergence checkpoint. On the same seed, the schedule also reaches a lower $\operatorname{tr}\operatorname{Cov}$ than the fixed-lr run at the matching 7×10^5 checkpoint (0.128 vs. 0.162); the 5-seed mean at 1×10^6 carries enough between-seed spread that it should be read mainly as evidence of stability, not as a precise depth estimate (see main text). $\operatorname{tr}\operatorname{Cov}$, velocity error, $\|\text{mean} - x^i\|$, and the training loss all decrease together throughout, identifying P1–P3 as one phenomenon driven by loss $\to 0$ (the exact minimiser has loss 0).

A literature-standard memorisation ratio confirms Corollary 1. The diagnostics above are all aggregate (trace of a sample covariance, or distances of the sample *mean*). As a per-sample complement, we report the *memorisation ratio* of Yoon et al. (2023)[1]: a generated sample $\hat{x}$ is *memorised* if $\|\hat{x} - x^{(1)}\|^2 \leq c\,\|\hat{x} - x^{(2)}\|^2$ for $c = 1/9$, where $x^{(1)}, x^{(2)}$ are the nearest and second-nearest training points to $\hat{x}$. Figure 5 shows this ratio for the same conditional and unconditional EXP-1 runs as Figure 1 (5 seeds each): both rise from a common ≈ 0.11 baseline as training proceeds, but the conditional ratio reaches 0.650 ± 0.033 at 2×10^5 iterations versus 0.389 ± 0.032 unconditionally—direct, literature-standard confirmation that *both* flows memorise (Corollary 1), with conditional memorisation the stronger of the two, exactly as the single-atom vs. full-empirical-measure distinction predicts. The two ratios are not directly comparable in magnitude—the unconditional target is spread over N=200 candidate atoms rather than one, so the $c = 1/9$ dominance criterion is mechanically harder to satisfy there even at equally exact memorisation—so we read this only as a directional confirmation (both rise well above baseline; conditional rises further), not as a calibrated measure of *how much more* conditional memorisation occurs. On EXP-2 (Section 4.6), the same ratio reaches 0.83 ± 0.12 by 3×10^5 iterations (2 seeds), tracking the mode-coverage collapse there. Pushed further still, the extended learning-rate-scheduled run of Section 4.1 reaches 0.955 ± 0.020

[1]As used, e.g., by Buchanan et al. (2025).

by 10^6 iterations (5 seeds): 95% of individual samples are memorised by this point, not merely the sample mean.

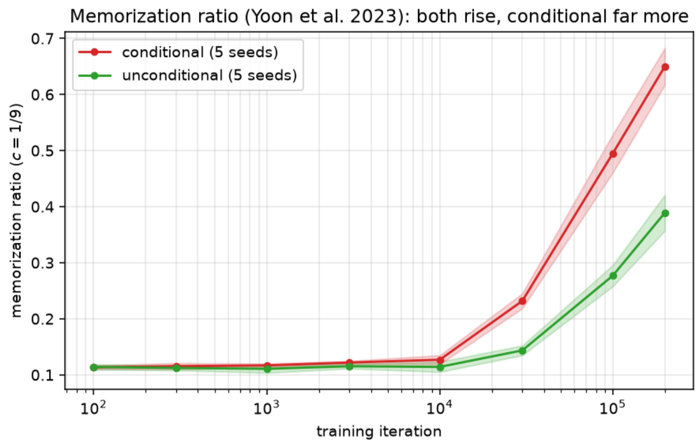


Figure 5: Memorisation ratio (5 seeds each) for the same conditional and unconditional EXP-1 runs as Figure 1. Both rise from the common ≈ 0.11 baseline of a well-calibrated sampler; the conditional ratio rises higher (0.650 ± 0.033 at 2×10^5) than the unconditional one (0.389 ± 0.032), consistent with single-atom vs. full-empirical-measure memorisation (Corollary 1).

Architecture and optimiser sensitivity. Every result above uses one fixed architecture (4-layer, width-128 MLP, Adam). Since Proposition 2 constrains only the *population* minimiser, not any parameterisation, the theory predicts collapse for *any* sufficiently expressive architecture; what it does not predict is how fast a given architecture and optimiser reach it within a fixed iteration budget. We re-ran EXP-1 (conditional, 2 seeds each) varying one factor at a time—width $\in \{32, 64, 256\}$ at depth 4, depth $\in \{2, 6\}$ at width 128, and the optimiser (SGD with momentum 0.9, lr $= 10^{-2}$, in place of Adam)—holding everything else, including the 2×10^5-iteration budget, fixed.

configuration	width	depth	optimiser	tr Cov at 2×10^5
baseline	128	4	Adam	0.396 ± 0.127
narrower	32	4	Adam	0.657 ± 0.058
	64	4	Adam	0.526 ± 0.058
wider	256	4	Adam	0.403 ± 0.079
shallower	128	2	Adam	0.860 ± 0.156
deeper	128	6	Adam	0.407 ± 0.127
SGD	128	4	SGD(m=0.9)	0.711 ± 0.193

Table 2: Architecture/optimiser ablation on EXP-1 (2 seeds each), same 2×10^5-iteration budget as Table 1. Collapse speed increases with capacity but saturates beyond roughly the baseline's size (width 256 and depth 6 both land within seed noise of the width-128/depth-4 baseline, despite reaching a strictly lower training loss—0.46 vs. 0.59 for width 256—so more capacity buys a better loss without a proportionally deeper collapse at this budget); under-capacity architectures (width 32, depth 2) collapse markedly less within the same budget; and the optimiser matters substantially, with SGD reaching only about half the collapse Adam does here.

None of these variants contradicts Proposition 2: all seven configurations collapse monotonically from the same $\approx \operatorname{tr}\Sigma_{\text{post}}$ starting point (not shown), and none plateaus above the baseline's level by more than a factor of ~ 2—this is exactly the qualitative picture the theory predicts (every sufficiently expressive architecture converges toward the same population optimum) together with the optimisation-gap caveat of Section 5 (how close a given budget gets you depends on the architecture and optimiser). We did not extend this sweep to the Lipschitz diagnostic of Section 4.5 or to EXP-2/EXP-3; doing so is left to future work.

4.2 Adversarial pairing: collapse does not require a real posterior

Proposition 2 never uses the forward operator A or σ_{obs}: it only needs the labels $y^1, \dots, y^N$ to be pairwise distinct. The theory therefore predicts that collapse is unchanged if Y is *randomly permuted relative to* X before training, i.e. if (x^i, y^i) is no longer the true generative pair—an identification mechanism, not evidence that the model discovered real posterior structure. This mirrors the "Adversarial Pairings" experiment of Reu et al. (2025) (shuffling the targets of ground-truth OT couplings on CelebA to test unconditional memorisation), adapted here to the conditional case, where the prediction is exact rather than a generic-position argument (cf. Remark 1).

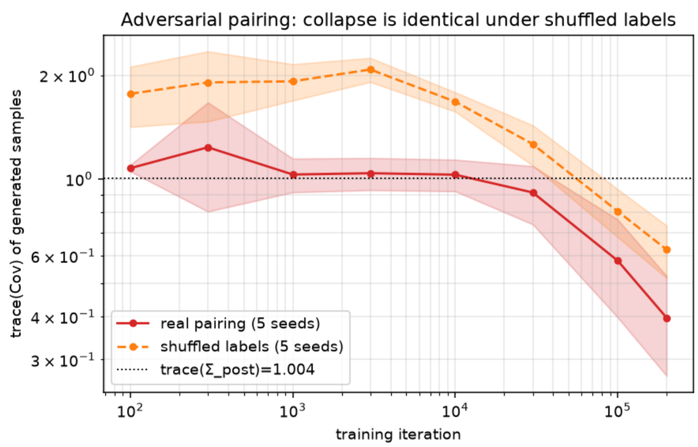


Figure 6: $\operatorname{tr}\operatorname{Cov}$ under the real EXP-1 pairing (5 seeds, as in Fig. 1) vs. the same architecture and schedule trained on Y randomly permuted relative to X (5 seeds). Both collapse monotonically well below $\operatorname{tr}\Sigma_{\text{post}}$; the shuffled run lags the real pairing at 2×10^5 iterations (0.625 ± 0.108 vs. 0.396 ± 0.127)—an optimisation-gap difference, not a failure of Proposition 2, whose population target is 0 either way.

We re-run the EXP-1 architecture and training schedule unchanged (5 seeds), permuting Y with an independent random permutation per seed before training (configs/exp1_adversarial_shuffle.yaml). Figure 6 confirms the qualitative prediction: $\operatorname{tr}\operatorname{Cov}$ collapses under shuffled labels just as it does under the real pairing, from $\gtrsim 1$ down to 0.625 ± 0.108 at 2×10^5 iterations (vs. 0.396 ± 0.127 for the real pairing)—the gap is an optimisation-difficulty difference at a fixed finite iteration budget (a random x-y pairing is a harder target function to fit than the smooth linear-Gaussian one), not a difference in the population optimum, which Proposition 2 fixes at 0 regardless of how Y was assigned.

The sharper diagnostic is Figure 7: $\|\text{mean} - \mu_{\text{post}}(y^i)\|$, the distance to y^i's *true* analytic posterior mean (still well defined and computable from A—it is simply now unrelated to the x^i the label was shuffled onto), does not shrink; it *grows* throughout training, crossing above $\|\text{mean} - x^i\|$ by 10^5 iterations. The generated law tracks the arbitrary label-to-example assignment, not any statistical relationship between x and y—the same conclusion Corollary 1 draws from the exact theory, now confirmed even when the labels carry no real information about x at all.

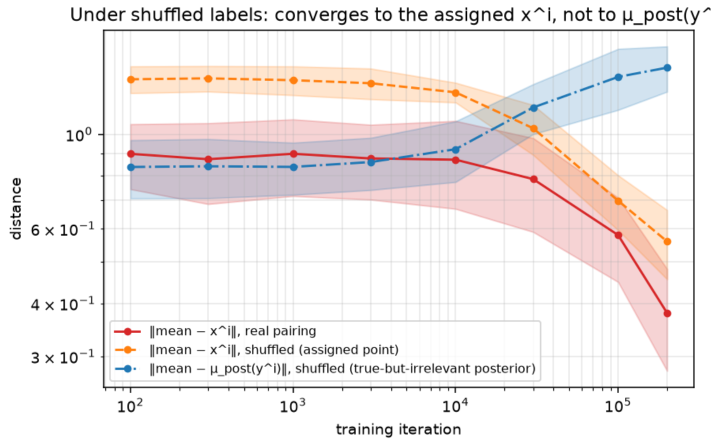


Figure 7: Under shuffled labels, the generated mean converges to the *assigned* training point x^i ($\|\text{mean} - x^i\| : 1.35 \pm 0.10 \to 0.559 \pm 0.104$) while moving *away* from the true analytic posterior mean of that y^i ($\|\text{mean} - \mu_{\text{post}}(y^i)\| : 0.84 \pm 0.13 \to 1.437 \pm 0.176$)—the two curves cross between 10^4 and 3×10^4 iterations. Under the real pairing the corresponding gap is far smaller ($\|\text{mean} - \mu_{\text{post}}\| = 0.656 \pm 0.177$ at 2×10^5, Table 1), since there x^i and $\mu_{\text{post}}(y^i)$ are close by construction.

4.3 Scope-marked sweeps (P5, P6, d/k; empirical)

The following depend on N and σ_{obs} at fixed architecture and are *not* determined by the population theory (Section 3.6); we report them as empirical findings using the collapse ratio $\operatorname{tr}\operatorname{Cov} / \operatorname{tr}\Sigma_{\text{post}}$ at 2×10^5 iterations.

N	50	200	1000	5000
ratio	0.049	0.389	0.912	0.990
$\pm$	0.038	0.123	0.103	0.031

(a) P6: collapse vs. dataset size.

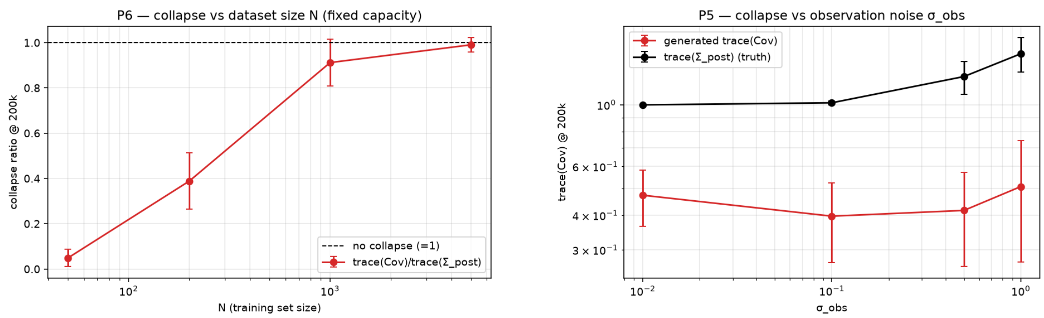
(d, k)	$(2, 1)$	$(10, 1)$	$(10, 10)$
ratio	0.390	0.317	0.0065
$\pm$	0.127	0.033	0.0024

(b) Collapse vs. dimension/observation.

Table 3: 5-seed sweeps (empirical). P6 is monotone and stable across seeds: small N collapses almost completely, N=5000 barely collapses ($\approx$ true posterior)—a representation/optimisation statement, not a contradiction of Proposition 2. Larger k (a more informative label) sharpens the identifier and deepens collapse, consistent with the identification mechanism.

σ_{obs}	0.01	0.1	0.5	1.0
$\operatorname{tr}\Sigma_{\text{post}}$	1.000	1.018	1.266	1.531
generated $\operatorname{tr}\operatorname{Cov}$	0.472 ± 0.108	0.396 ± 0.127	0.416 ± 0.155	0.507 ± 0.236
collapse ratio	0.472	0.390	0.328	0.331

Table 4: P5 (empirical), 5 seeds. The between-σ_{obs} spread in the ratio (0.33–0.47, range ~ 0.14) is *smaller* than the between-seed std at each point (0.11–0.24): the sweep is flat within seed noise. This is consistent with an identification-through-y mechanism that operates for every $\sigma_{\text{obs}} > 0$; the theory implies no monotonicity, so flatness is not a falsification.



(a) P6: collapse ratio vs. dataset size N.

(b) P5: collapse ratio vs. observation noise σ_{obs}.

Figure 8: Scope-marked sweeps (5 seeds, error bars over seeds). Collapse deepens monotonically as N shrinks (a); it is flat within seed noise across σ_{obs} (b). Neither is determined by the population theory; both are reported as empirical findings (Section 3.6).

4.4 Label smoothing tracks the kernel reference (P7)

This is our strongest empirical confirmation of the theory. The correct reference is Cov_h (Theorem 1, Proposition 5), *not* Σ_{post}. Figure 9 summarises the result: the generated variance tracks the exact kernel target Cov_h across every bandwidth (panel a), the ratio-to-kernel sits in a tight band around 1 for $h \geq 0.05$ (panel b), and yet the effective number of atoms n_{eff} stays far below N=200 (panel c), so the law remains atomic even where the second moment matches. Table 5 evaluates the trained checkpoints against the exact kernel moments over 20 conditions (M=1000), across five seeds.

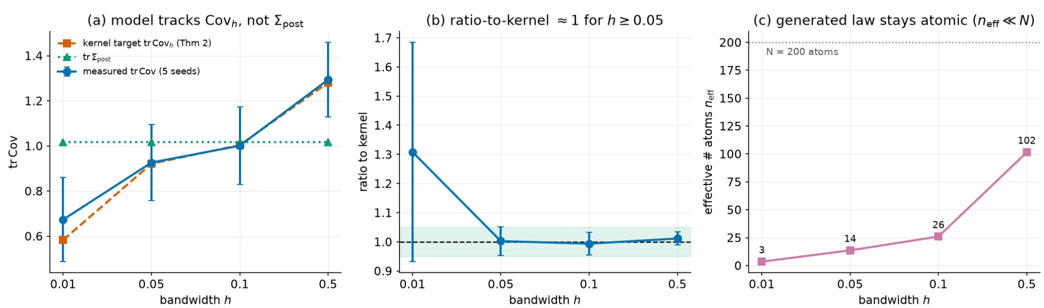


Figure 9: P7 headline (5 seeds, error bars over seeds). **(a)** The generated $\mathrm{tr}\,\mathrm{Cov}$ follows the kernel target $\mathrm{tr}\,\mathrm{Cov}_h$ (dashed) and crosses $\mathrm{tr}\,\Sigma_{\text{post}}$ (dotted, flat) only incidentally near h=0.1. **(b)** The ratio-to-kernel lies within the shaded $[0.95, 1.05]$ band for $h \geq 0.05$. **(c)** $n_{\text{eff}} \ll N$: the endpoint law is a sparse mixture of training atoms (Proposition 6), not a continuous posterior.

h	Cov_h (Thm 1)	Σ_{post}	generated tr Cov	**ratio-to-kernel**	ratio-to-post	n_{eff}/200
0.01	0.583	1.018	0.673 ± 0.187	$1.308 \pm 0.376^{\dagger}$	0.66	3.5
0.05	0.921	1.018	0.926 ± 0.169	$\mathbf{1.002 \pm 0.050}$	0.91	13.6
0.10	1.004	1.018	1.001 ± 0.173	$\mathbf{0.994 \pm 0.039}$	0.98	26.0
0.50	1.281	1.018	1.294 ± 0.166	$\mathbf{1.011 \pm 0.022}$	1.27	101.5

Table 5: P7 kernel reference, 5 seeds. For every $h \geq 0.05$ the generated variance tracks the exact kernel target Cov_h to within seed noise (ratio-to-kernel ≈ 1.00, std ≤ 0.05)—a verification of the *endpoint law*, stronger than a velocity-field check. †At h=0.01 ($n_{\text{eff}} \approx 3.5$, single-atom regime) Cov_h is near-degenerate for many conditions, so the per-condition ratio is unstable and high; the ratio of mean traces $0.673/0.583$=1.15 is closer to 1.

A natural but mistaken reading of Table 5 is that $h \approx 0.1$ "perfectly restores" the posterior variance: $\mathrm{tr}\,\mathrm{Cov}$ merely crosses $\Sigma_{\text{post}} \approx 1.02$ near h=0.1 because for this problem $\Sigma_{\text{post}} \approx \mathrm{Cov}_h|_{h=0.1} \approx$

1.00—a coincidence. The correct target is Cov_h, which the model attains at *every* h, not only h=0.1. Figure 10 and Table 6 confirm the field itself: the learned v_θ matches the kernel minimiser (3) far better than the single-atom field ($\star$), and the gap widens with h, exactly as the mixture structure predicts.

h	rel. error vs. **kernel** (3)	rel. error vs. ($\star$)	TV of atom mixture
0.01	$\mathbf{0.318 \pm 0.030}$	0.599 ± 0.104	0.31 ± 0.18
0.05	$\mathbf{0.195 \pm 0.019}$	0.683 ± 0.103	0.22 ± 0.02
0.10	$\mathbf{0.162 \pm 0.021}$	0.703 ± 0.092	0.16 ± 0.04
0.50	$\mathbf{0.164 \pm 0.025}$	0.791 ± 0.070	0.16 ± 0.02

Table 6: Velocity-field match (5 seeds). v_θ is 2–4× closer to the kernel field than to the collapsed field at every h. Assigning generated samples to the nearest atom recovers the weights $p_i^{(h)} \propto K_h$ with total variation 0.16–0.31. At h=0.5 the generated variance 1.29 *overshoots* Σ_{post}=1.02, matching the $+h^2\|J\|_F^2$ term of Proposition 7 ($\|J\|_F^2 = 0.4031$, exact).

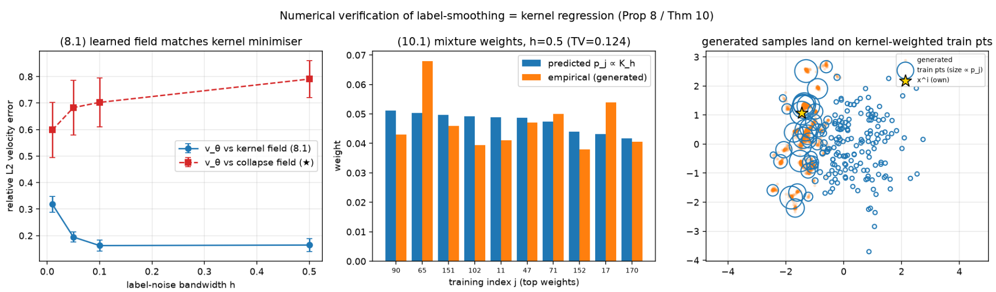


Figure 10: Kernel-theory verification: the learned field matches the kernel minimiser (3) (left); the predicted atom weights $p_i^{(h)} \propto K_h$ match the empirical nearest-atom assignment (centre); and, for one condition y, generated samples (orange) land on the training points weighted most heavily by the kernel (blue circles, radius $\propto p_i^{(h)}$), confirming Theorem 1's mixture-over-atoms endpoint law directly in sample space (right).

Atomicity caveat (empirical confirmation of Proposition 6). Even at the h that best matches Σ_{post}, the generated law remains atomic. The effective number of atoms is only $n_{\text{eff}} \approx 26/200$ at h=0.1 and $\approx 102/200$ at h=0.5 (Table 5). Directly measuring the discrepancy to the analytic posterior (Section 4.5) shows the MMD decreasing with h but never reaching zero at any h. Matching variance is not recovering the posterior.

Interpolant noise does not help (Proposition 9). Table 7 adds stochastic interpolant noise. With the *corrected* target (8) (verified by four closed-form checks), the generated variance is unchanged from that obtained with the mis-specified $X_1 - X_0$ target, and remains far below the posterior—exactly as (9) predicts: the endpoint law is δ_{x^i} for every σ, independently even of whether the target is correct. Only label noise, acting on the *label* factor, can change p_1 (Proposition 10).

interpolant σ	0.1	0.3
tr Cov, old target $X_1 - X_0$ (2 seeds)	0.372 ± 0.140	0.429 ± 0.121
tr Cov, corrected target (8) (5 seeds)	$\mathbf{0.367 \pm 0.146}$	$\mathbf{0.383 \pm 0.131}$
mean bias	0.68	0.66

Table 7: Interpolant-noise contrast. Correcting the target leaves the result unchanged within seed noise—a confirmation of Proposition 9(c), not a failure.

4.5 ADDITIONAL VERIFICATION (GAP TO OPTIMUM, LIPSCHITZ, POSTERIOR DISTANCE)

Three checkpoint-only measurements corroborate the theory. (i) *Optimality gap.* The Monte-Carlo gap $L(v_\theta) - L(v_h^\star)$, with $L(v_h^\star) = \mathbb{E}[\mathrm{Var}(U \mid X_t, t, \widetilde{Y})]$ computed from the exact minimiser, is non-negative at every checkpoint and decreases monotonically toward 0 at every h (e.g. h=0.1: $1.14 \to 0.22 \to 0.102$ at $10^2/3{\times}10^4/2{\times}10^5$ steps), consistent with ratio-to-kernel $\to$ 1. The irreducible error $L(v_h^\star)$ increases with h ($0 \to 1.88$): the price of smoothing (Figure 11).

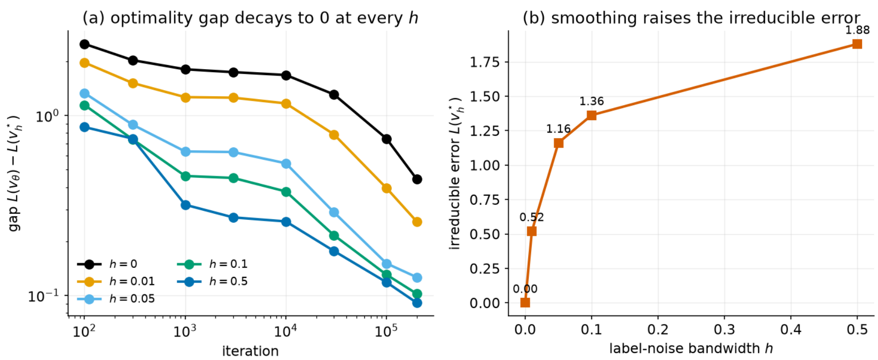


Figure 11: Optimality gap $L(v_\theta) - L(v_h^\star)$ decays monotonically to 0 at every h (left); smoothing raises the irreducible error $L(v_h^\star)$ (right).

(ii) *Lipschitz floor.* The empirical $\mathrm{Lip}_x\, v_\theta$ grows from 7.1 at t=0.5 to 65.7 at t=0.99; the Corollary 5 floor $d/(3L) \approx 0.010$ is $\sim 50\times$ below the observed plateau 0.48, so this particular representation-capacity argument does *not* explain the plateau—by elimination the residual gap is not a Lipschitz-capacity effect, though we do not have a positive theory of what it is instead (Section 5) (Figure 12).

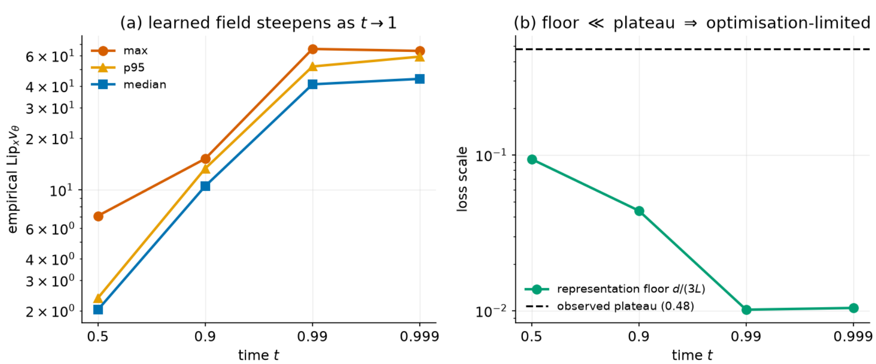


Figure 12: The empirical Lipschitz constant of v_θ blows up as $t \to 1$ (left); the representation floor $d/(3L)$ is $\sim 50\times$ below the observed loss plateau (right), so the plateau is optimisation-limited.

(iii) *Posterior distance.* The MMD to the analytic posterior $\mathcal{N}(\mu_{\mathrm{post}}, \Sigma_{\mathrm{post}})$ falls with h ($0.256 \to 0.012$) but never reaches 0, and the Sinkhorn distance plateaus near 2.7—a direct numerical witness of the h-independent floor (6) (Figure 13).

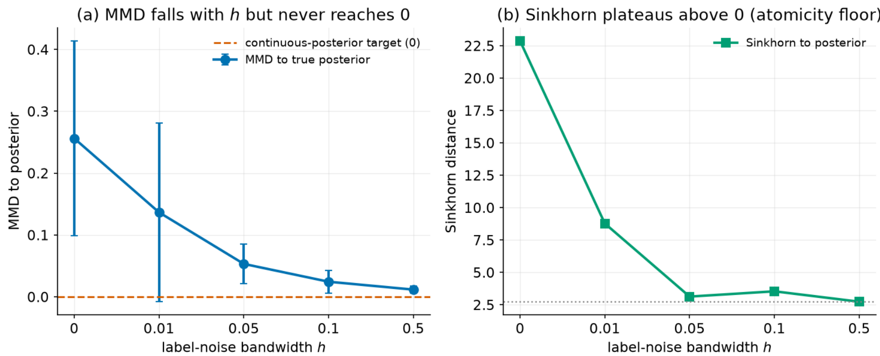


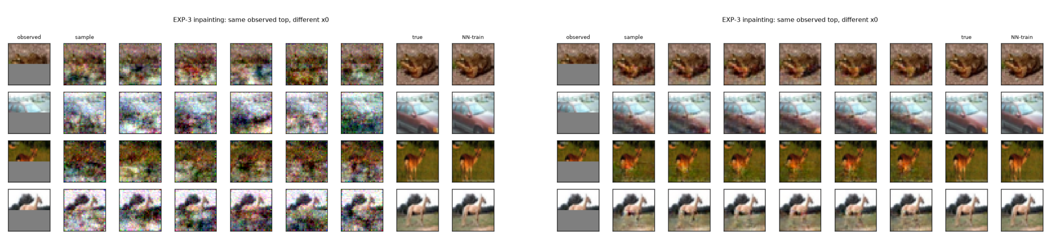
Figure 13: MMD (left) and Sinkhorn (right) to the analytic posterior fall with h but never reach 0—the h-independent atomicity floor of Proposition 6.

4.6 EXP-2 AND EXP-3: SELECTIVE MEMORISATION BEYOND THE GAUSSIAN CASE

EXP-2 (Gaussian mixture). A 4-mode GMM prior with a rank-deflating linear observation gives a genuinely *bimodal* analytic posterior. As the loss descends (Fig. 15, 2 seeds), mode coverage falls $1.00 \to 0.72$ (toward 0.5, i.e. one of the two modes) and the MMD to the true posterior grows from ~ 0.04 to ~ 0.25; up to $\sim 10^4$ iterations coverage is a full 1.0 and the MMD is near zero—the model is a calibrated posterior sampler—before overtraining drives it onto x^i in a single mode and abandons the other. This is selective memorisation, and its onset at $\sim 10^4$ iterations mirrors EXP-1.

EXP-3 (MNIST inpainting). Masking the lower half of 32×32 MNIST ($N{=}500$) with a small U-Net, we draw 16 completions per condition. Over *3 seeds*, the inpainted-region pixel variance drops $\sim 100\times$ by iteration 15000 ($0.137 \pm 0.021 \to (1.17 \pm 0.12) \times 10^{-3}$) while the distance to the nearest training image falls to $(0.77 \pm 0.26) \times 10^{-3}$ and the observed-region reconstruction stays small, $(1.7 \pm 1.3) \times 10^{-4}$ (larger and noisier across seeds than the single-seed value reported in an earlier draft of this experiment, 1.9×10^{-5}, but still three orders of magnitude below the pixel-variance collapse). Qualitatively, diverse completions at iteration 200 sharpen progressively and become identical by iteration 15000, coinciding with the nearest training image (Fig. 18)—the same collapse on real image data, reached earlier because the loss converges deeper here.

A harder dataset: CIFAR-10 inpainting. To check this is not an artefact of MNIST's simplicity, we repeat EXP-3 verbatim (same architecture, $N{=}500$, 15000 iterations) on CIFAR-10—natural RGB images with far more texture and per-pixel variability than handwritten digits (one seed; see Section 5). The same collapse occurs: inpainted-region pixel variance falls from 0.142 to 0.0087 ($\sim 16\times$) and the nearest-training-image distance falls from 0.057 to 0.0040, while completions visibly sharpen from diverse noisy blobs at iteration 200 to samples indistinguishable from each other and from the memorised training image by iteration 15000 (Figure 14).



(a) iter 200 (b) iter 15000

Figure 14: EXP-3 on CIFAR-10 (one seed), same protocol as Figure 18. At iteration 200, completions of the masked bottom half are diverse and noisy; by iteration 15000 they are visually identical to each other and to the nearest training image (rightmost two columns), confirming the collapse mechanism is not specific to MNIST's simplicity.

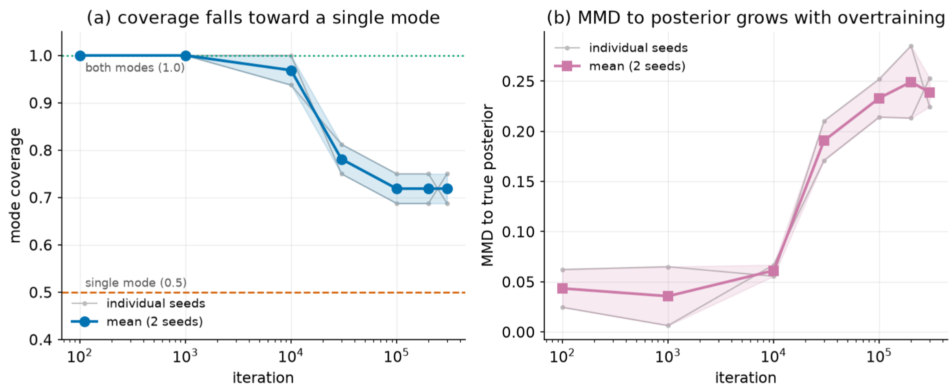


Figure 15: EXP-2 (GMM), mean of 2 seeds with per-seed traces and a ±std band. **(a)** Mode coverage holds at 1.0 (both posterior modes) until $\sim 10^4$ iterations, then falls toward the single-mode line 0.5. **(b)** The MMD to the true bimodal posterior is near zero in the calibrated phase and grows as overtraining concentrates the samples on one mode. Onset matches EXP-1.

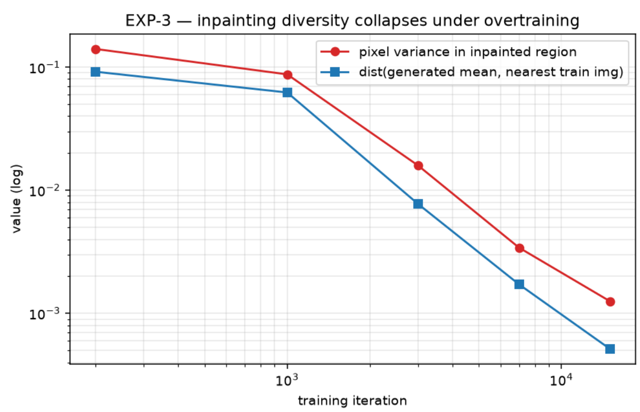


Figure 16: EXP-3 (MNIST inpainting), quantitative collapse: the inpaint-region pixel variance and the distance to the nearest training image both fall by orders of magnitude as the loss descends, while the observed-region reconstruction stays exact.

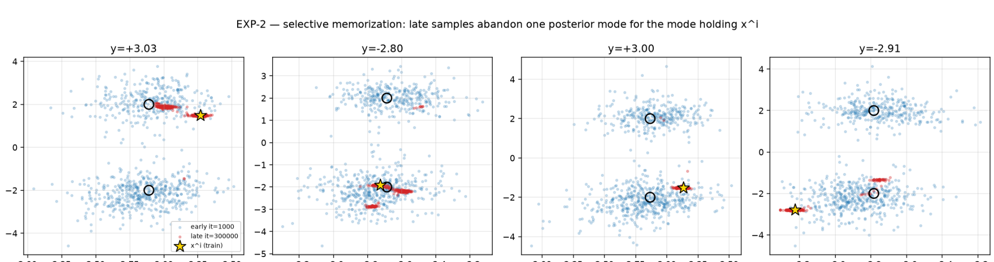
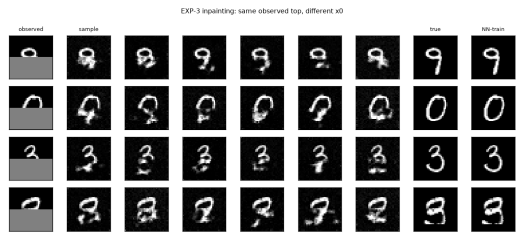
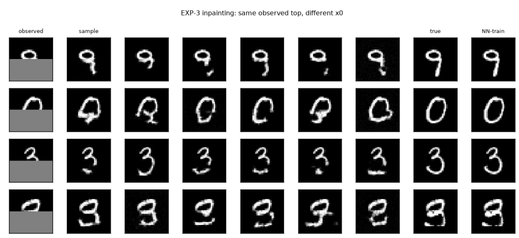


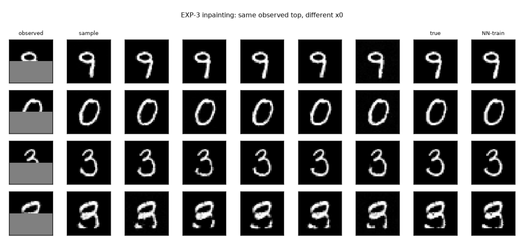
Figure 17: EXP-2 qualitative collapse. Generated samples concentrate on the posterior mode containing the memorised x^i and abandon the other, even though the analytic posterior for this observation is bimodal.



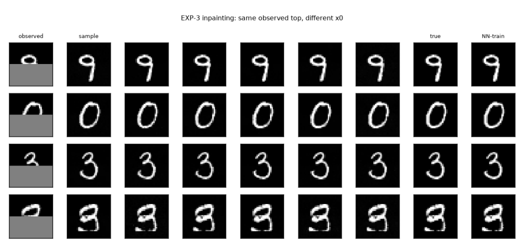
(a) iter 200



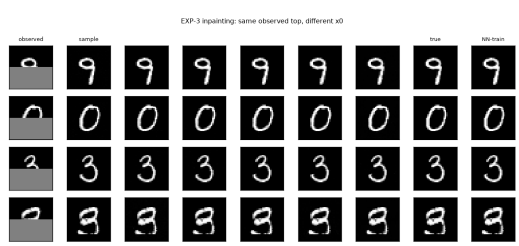
(b) iter 1000



(c) iter 3000



(d) iter 7000



(e) iter 15000

Figure 18: EXP-3 collapse trajectory (MNIST inpainting). Each grid shows completions of a fixed masked digit from different source samples x_0. Early in training (iter 200) the completions are diverse—a calibrated conditional sampler; as the loss descends they sharpen progressively (iter 1000–7000) and, by iter 15000, are identical to one another and equal to the nearest training image (inpaint-region pixel variance $0.137 \rightarrow 0.0012$ averaged over 3 seeds, a $\sim 100\times$ drop). This is the same loss $\rightarrow$ 0-paced collapse as EXP-1, on real image data, and it reproduces on CIFAR-10 too (Figure 14).

5 LIMITATIONS

The theory does not quantitatively explain the residual optimisation gap. A finite network trained by SGD approaches the population minimiser only up to an optimisation gap, which is why hard-conditioning collapse is *optimisation-paced* and only partial within a finite budget (for $h > 0$ the model does reach the population optimum Cov_h). Corollary 5's Lipschitz-capped representation floor was our candidate explanation for why the $h = 0$ plateau stalls above zero, but Section 4.5 shows it does *not* bind here—the floor is $\sim 50\times$ below the observed plateau. At a *fixed* learning rate, extreme overtraining eventually diverges near 10^6 iterations, since the collapsed target's Lipschitz constant grows as $t \to 1$ and a constant step size becomes too large once the loss is small; a cosine learning-rate decay removes this divergence and lets training continue collapsing further without blowing up (Table 1, Figure 4), confirming that the plateau is a *transient*, schedule-sensitive optimisation phenomenon rather than a hard floor. We therefore cannot currently say *why* training stalls at the specific rate it does at a fixed learning rate, or predict the schedule that minimises iterations-to-collapse; characterising this optimisation gap quantitatively—as opposed to merely bounding it away from a representation explanation and showing empirically that it is schedule-sensitive—remains an open problem this paper does not solve.

All experiments are small-scale. EXP-1/EXP-2 are $d \leq 10$ synthetic linear-Gaussian/Gaussian-mixture problems, and EXP-3 (now 3 seeds on MNIST, plus one seed on CIFAR-10 as a harder, more textured dataset; Section 4.6) still uses only 32×32 images, N=500, and one small U-Net—we do not validate the theory on larger images, other modalities, or the U-Net/transformer backbones used in production diffusion and flow models. We expect the population-level mechanism to persist at scale (it is architecture-agnostic by construction, since Proposition 2 constrains only the population minimiser, not any particular parameterisation), and the CIFAR-10 replication is one data point in that direction, but a systematic study across resolutions, modalities, and production-scale backbones remains future work; so does a mask-type and N sweep for EXP-3, and extending CIFAR-10 to multiple seeds.

Limited architecture and optimiser sensitivity study. All headline EXP-1/EXP-2 results use one fixed architecture (4-layer, width-128 MLP with SiLU) and optimiser (Adam, lr $= 10^{-3}$). Section 4.1 reports a small width/depth/optimiser sweep (2 seeds per configuration) confirming that collapse is not an artefact of this specific choice, but that sweep is limited to $\operatorname{tr}\mathrm{Cov}$ at a single iteration budget on EXP-1; we do not study how the empirical Lipschitz constant of Section 4.5 depends on width or depth (Section 3.4's representation-floor argument is sensitive to exactly this), nor do we repeat the sweep on EXP-2 or EXP-3, nor sweep the optimiser's learning rate or the SGD momentum coefficient.

The paper is diagnostic, not prescriptive. We characterise why collapse happens and why two natural remedies (label smoothing, interpolant noise) do or do not work, but we do not propose or evaluate a working fix: the atomicity obstruction (Proposition 6) means the label-smoothing "remedy" matches moments but never recovers a continuous posterior, and we offer no alternative that does.

Other caveats. The sweeps P5 and P6 are empirical findings the population theory does not determine and must not be read as prediction matches. The random seed used in EXP-1/EXP-2 controls both the problem instance and the training run jointly (Section 4). Checkpoints are regenerable and not tracked in version control, so reproducing the checkpoint-based tables from a clean clone requires retraining the $p7y$ runs (deterministic per seed); all sweep tables reproduce directly from the tracked metrics.

6 CONCLUSION

Conditional collapse in flow matching is the zero-bandwidth limit of a kernel-weighted empirical conditional flow. Under hard conditioning with identifying labels the exact population minimiser has zero conditional variance; label smoothing turns this into exact kernel regression over the training atoms, with a parameter-free moment reference that a trained network tracks to within seed noise across five seeds. The two apparent remedies are provably different—label noise moves the population endpoint, interpolant noise cannot—and even the working remedy only matches moments, bounded away from the true posterior by an h-independent Wasserstein floor. The same collapse holds on a bimodal GMM posterior, on MNIST and CIFAR-10 inpainting, across a range of MLP widths/depths and both Adam and SGD, and—confirmed by a literature-standard memorisation ratio—even when the training labels are shuffled and carry no real information about x at all, exactly as the identification mechanism predicts.

REPRODUCIBILITY STATEMENT

All numbers reported are produced by code included in the supplementary material (anonymized for review). Theory checks: `test_kernel_theory.py` (reference table to $< 0.05\%$), `analyze_p7_kernel.py`, `verify_kernel_theory.py`, `verify_prop17.py`. Experiments: `run_exp1.sh`, `run_sweeps.py` (+`--gpus N`), `train_exp2`, `train_exp3`, each followed by the matching analysis script. Proofs of all load-bearing results are given in full in Section 3 and Appendix A; every stated assumption (distinct labels, smoothness of the weight functions, the linear-Gaussian instance) is made explicit at the point it is used.

ETHICS STATEMENT

This work trains small models on synthetic linear-Gaussian and Gaussian-mixture data and on the public MNIST dataset. No human subjects, personally identifiable information, or other sensitive data are used or generated. The paper's contribution is a precise theoretical and empirical account of an existing memorisation phenomenon in conditional generative models, which bears directly on the data-privacy and copyright concerns around deployed diffusion/flow models raised in Section 1; by characterising the mechanism exactly, we aim to help downstream work design principled mitigations rather than to facilitate memorisation. We are not aware of any other ethical concerns raised by this submission.

AI USE STATEMENT

We used a generative-AI coding and writing assistant (Claude, Anthropic) throughout this project's development: to draft and revise manuscript text, including the conversion of the manuscript to the ICLR format and multiple rounds of copy-editing; to expand several of the proofs in Section 3 and Appendix A from a theory memorandum into the propositions and proofs given here; to write the training, evaluation, and analysis code in the released repository; and to run and analyse several of the reported experiments, including the adversarial-pairing ablation (Section 4.2) and the memorisation-ratio diagnostic (Section 4.1). We did not use generative AI to select, fabricate, or embellish experimental results: every number reported is produced by the released code from actual training runs. All AI-drafted proofs were checked line-by-line against the underlying mathematics by the authors, and all AI-assisted code and analysis outputs were independently verified by re-running the experiments and cross-checking the reported statistics against the raw output files. We take responsibility for the final content of this work, including all text, claims, proofs, and code.

REFERENCES

Giulio Biroli and Marc Mézard. Generative diffusion in very large dimensions. *Journal of Statistical Mechanics: Theory and Experiment*, 2023(9):093402, 2023. arXiv:2306.03518.

Sam Buchanan, Druv Pai, Yi Ma, and Valentin De Bortoli. On the edge of memorization in diffusion models. *arXiv preprint arXiv:2508.17689*, 2025. Unconditional memorization/generalization phase transition as a function of denoiser capacity M relative to N, via a Gaussian-mixture-denoiser crossover-point theory; complementary to our label/bandwidth axis.

Nicholas Carlini, Jamie Hayes, Milad Nasr, Matthew Jagielski, Vikash Sehwag, Florian Tramèr, Borja Balle, Daphne Ippolito, and Eric Wallace. Extracting training data from diffusion models. In *32nd USENIX Security Symposium*, 2023. arXiv:2301.13188; verbatim training-image extraction from deployed diffusion models.

Agnimitra Dasgupta, Ali Fardisi, Mehrnegar Aminy, Brianna Binder, Bryan Shaddy, and Assad Oberai. Solving physics-constrained inverse problems with conditional flow matching. *arXiv preprint arXiv:2603.14135*, 2026. Source of the observed variance-collapse / selective-memorization phenomenon.

Xiangming Gu, Chao Du, Tianyu Pang, Chongxuan Li, Min Lin, and Ye Wang. On memorization in diffusion models. *arXiv preprint arXiv:2310.02664*, 2023.

Zahra Kadkhodaie, Florentin Guth, Eero P. Simoncelli, and Stéphane Mallat. Generalization in diffusion models arises from geometry-adaptive harmonic representations. In *International Conference on Learning Representations (ICLR)*, 2024. Oral.

Mason Kamb and Surya Ganguli. An analytic theory of creativity in convolutional diffusion models. *arXiv preprint arXiv:2412.20292*, 2024. ICML 2025.

Jakiw Pidstrigach. Score-based generative models detect manifolds. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2022. arXiv:2206.01018.

Teodora Reu, Sixtine Dromigny, Michael Bronstein, and Francisco Vargas. Gradient variance reveals failure modes in flow-based generative models. *arXiv preprint arXiv:2510.18118*, 2025. Unconditional memorization via the deterministic interpolant; the resample-x_0 mechanism.

Gowthami Somepalli, Vasu Singla, Micah Goldblum, Jonas Geiping, and Tom Goldstein. Diffusion art or digital forgery? investigating data replication in diffusion models. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2023. arXiv:2212.03860; image-retrieval-based empirical detection of training-data replication.

TaeHo Yoon, Joo Young Choi, Sehyun Kwon, and Ernest K. Ryu. Diffusion probabilistic models generalize when they fail to memorize. In *ICML 2023 Workshop on Structured Probabilistic Inference & Generative Modeling*, 2023. Origin of the nearest/second-nearest-neighbour memorisation ratio, $c = 1/9$.

A Deferred proofs

Lemma 4 (Change of variables)**.** *Fix i and $t < 1$. The map $X_0 \mapsto X_t = (1-t)X_0 + tx^i$ is invertible affine with $X_0 = (X_t - tx^i)/(1-t)$, so $\sigma(X_t \mid I = i) = \sigma(X_0 \mid I = i)$ and the conditional density is $p(x \mid I = i, t) = (1-t)^{-d}\pi_0\big(\frac{x-tx^i}{1-t}\big)$. The Jacobian $(1-t)^{-d}$ is i-independent and cancels in every posterior over I. Because π_0 is Gaussian, $p(x \mid I = i, t) > 0$ for every x and $t < 1$.*

Lemma 5 (Well-posedness)**.** *Let $w_1, \dots, w_N : \mathbb{R}^d \times [0,1) \to [0,1]$ be C^∞ with $\sum_i w_i \equiv 1$ and $v(x,t) = \sum_i w_i(x,t)(x^i - x)/(1-t)$. Then v is C^∞ hence locally Lipschitz on $\mathbb{R}^d \times [0,1)$, the ODE $\dot{x}_t = v(x_t, t)$ has a unique solution on $[0,1)$ for every initial condition, and with $M = \max_i |x^i|$, $|x_t| + M \le (|x_0| + M)/(1-t)$. Consequently the continuity equation has a unique measure solution and endpoint laws are the weak limits $p_1 = \lim_{t\uparrow 1} p_t$.*

Both $v^\star_{\text{unc}}$ and $v^\star_h$ have this form with smooth weights, so Lemma 5 applies and no "assume the flow is well defined" hypothesis is needed. We now give the remaining proofs in full.

Proof of Proposition 3. Equation (2)'s $q_i(x,t)$ is Bayes' rule applied to Lemma 4's density $p(x \mid I = i, t) = (1-t)^{-d}\pi_0\big(\frac{x-tx^i}{1-t}\big)$ under the uniform prior on I: the Jacobian $(1-t)^{-d}$ and the prior $1/N$ do not depend on i and cancel, leaving $q_i(x,t) \propto \pi_0\big(\frac{x-tx^i}{1-t}\big)$. Given $I = i$, Proposition 2(a)'s argument gives $U = (x^i - x)/(1-t)$; averaging over the posterior $q_i(x,t)$ and substituting into Proposition 1 gives $v^\star_{\text{unc}}$. For the endpoint law: by construction $v^\star_{\text{unc}}(x,t) = \mathbb{E}[\dot{X}_t \mid X_t = x, t]$, which is exactly the velocity field appearing in the continuity equation for the marginal law of the interpolation. By Lemma 5 the flow is well posed on $[0,1)$ and transports $p_0 = \pi_0$ to $p_t = \mathrm{Law}(X_t)$ for every $t < 1$; since $X_t \to X_1 \sim \widehat{\rho}_X$ almost surely, $p_t \Rightarrow \widehat{\rho}_X$, giving $p_1^{\text{unc}} = \widehat{\rho}_X$.

For $\inf_v L_{\text{unc}} > 0$: fix $t < 1$ and x. By the remark after Lemma 4, π_0 is everywhere positive, so $q_i(x,t) > 0$ for every i. Conditionally on $(X_t = x, t, I = i)$ the target is the deterministic vector

$u_i = (x^i - x)/(1-t)$, and $u_i \neq u_j$ whenever $x^i \neq x^j$. Hence the discrete law $\{q_i(x,t)\}_i$ places positive mass on at least two distinct values $u_i \neq u_j$ whenever some $x^i \neq x^j$, so $\mathrm{Var}(U \mid X_t = x, t) = \sum_i q_i(x,t)\,|u_i - \bar{u}(x,t)|^2 > 0$ at *every* (x,t) with $t < 1$, where $\bar{u}(x,t) = \sum_i q_i(x,t) u_i$. Integrating over (x,t) and applying Proposition 1, $\inf_v L_{\mathrm{unc}}(v) = \mathbb{E}[\mathrm{Var}(U \mid X_t, t)] > 0$. □

Proof of Proposition 7. Throughout, ρ_Y denotes the marginal density of Y and $s(y) := \nabla \log \rho_Y(y) \in \mathbb{R}^k$ its score at y, both well defined since ρ_Y is positive and C^2 near y.

Step 0: an exact identity via a tilted population measure. Define the h-tilted density of Y,

$$q_h(y') := \frac{K_h(y-y')\,\rho_Y(y')}{\int K_h(y-y'')\,\rho_Y(y'')\,dy''},$$

and, for a measurable f, $\mathbb{E}_h[f(X,Y)] := \mathbb{E}[K_h(y-Y)f(X,Y)]/\mathbb{E}[K_h(y-Y)]$, i.e. expectation under $Y \sim q_h$ with $X \mid Y$ unchanged. As $N \to \infty$ the empirical kernel-weighted moments $\bar{x}_h(y), \mathrm{Cov}_h(y)$ of Proposition 5 converge to their population counterparts $\mathbb{E}_h[X]$, $\mathrm{Cov}_h^{\mathrm{pop}}(y) := \mathrm{Cov}_h[X]$ (a law of large numbers for the weighted sums $\sum_j K_h(y-y^j)x^j$ and $\sum_j K_h(y-y^j)$ separately, then Slutsky's theorem for the ratio); the discrepancy between the finite-N empirical moments and this population limit is the $O_P((Nh^k)^{-1/2})$ term addressed in Step 3. It therefore suffices to show

$$\mathrm{Cov}_h^{\mathrm{pop}}(y) = \Sigma(y) + h^2 J(y)J(y)^\top + O(h^4). \tag{11}$$

By the law of total covariance applied to X given Y under the tilted law $Y \sim q_h$,

$$\mathrm{Cov}_h^{\mathrm{pop}}(y) = \underbrace{\mathbb{E}_h[\Sigma(Y)]}_{\text{within-group}} + \underbrace{\mathrm{Cov}_h[\mu(Y)]}_{\text{between-group}}, \tag{12}$$

which is an exact identity for every $h > 0$, since $\Sigma(Y) = \mathrm{Cov}(X \mid Y)$ and $\mu(Y) = \mathbb{E}[X \mid Y]$ by definition and the law of total covariance requires no approximation.

Step 1: the tilted law of Y, rescaled. Substitute $y' = y + hu$ ($dy' = h^k\,du$) and write $K_h(y - y') = h^{-k}\varphi(u)$ with $\varphi(u) = (2\pi)^{-k/2}e^{-|u|^2/2}$ the standard Gaussian density (immediate from the definition of K_h). Then

$$\int K_h(y-y'')\rho_Y(y'')\,dy'' = \int \varphi(u)\rho_Y(y+hu)\,du,$$

and by the C^2 Taylor expansion $\rho_Y(y+hu) = \rho_Y(y)\big[1 + h\,s(y)^\top u + O(h^2)\big]$ (uniformly over the region where φ has non-negligible mass, using C^2 regularity of ρ_Y near y), together with $\int \varphi(u)u\,du = 0$,

$$\int \varphi(u)\rho_Y(y+hu)\,du = \rho_Y(y)\big[1 + O(h^2)\big].$$

Hence the tilted law of $u := (Y-y)/h$ under q_h has density

$$\tilde{q}_h(u) = \frac{\varphi(u)\rho_Y(y+hu)}{\int \varphi(u')\rho_Y(y+hu')\,du'} = \varphi(u)\big[1 + h\,s(y)^\top u + O(h^2)\big]. \tag{13}$$

Using $\mathbb{E}_\varphi[u] = 0$, $\mathbb{E}_\varphi[uu^\top] = I_k$, $\mathbb{E}_\varphi[u_a u_b u_c] = 0$ (odd moments of a centred Gaussian vanish) and $\mathbb{E}_\varphi[u_a u_b u_c u_d] = \delta_{ab}\delta_{cd} + \delta_{ac}\delta_{bd} + \delta_{ad}\delta_{bc}$, (13) gives the moments

$$\mathbb{E}_{\tilde{q}_h}[u] = \mathbb{E}_\varphi[u] + h\,\mathbb{E}_\varphi\big[u\,(s^\top u)\big] + O(h^2) = h\,\mathbb{E}_\varphi[uu^\top]s + O(h^2) = h\,s(y) + O(h^2), \tag{14}$$

$$\mathbb{E}_{\tilde{q}_h}[uu^\top] = \mathbb{E}_\varphi[uu^\top] + h\,\mathbb{E}_\varphi\big[uu^\top(s^\top u)\big] + O(h^2) = I_k + 0 + O(h^2), \tag{15}$$

the vanishing middle term in (15) because $\mathbb{E}_\varphi[u_a u_b u_c] = 0$ for every a, b, c. Consequently

$$\mathrm{Cov}_{\tilde{q}_h}[u] = \mathbb{E}_{\tilde{q}_h}[uu^\top] - \mathbb{E}_{\tilde{q}_h}[u]\mathbb{E}_{\tilde{q}_h}[u]^\top = I_k + O(h^2) - h^2 s(y)s(y)^\top = I_k + O(h^2). \tag{16}$$

Step 2: within-group term. By C^1 smoothness of $\Sigma(\cdot)$, $\Sigma(y+hu) = \Sigma(y) + h\,[\nabla\Sigma(y)\cdot u] + O(h^2)$ (a matrix-valued first-order expansion, contracting the gradient 3-tensor with u), so by (13) and (14),

$$\mathbb{E}_h[\Sigma(Y)] = \mathbb{E}_{\tilde{q}_h}[\Sigma(y+hu)] = \Sigma(y) + h\,\big[\nabla\Sigma(y)\cdot\mathbb{E}_{\tilde{q}_h}[u]\big] + O(h^2) = \Sigma(y) + O(h^2),$$

since $\mathbb{E}_{\tilde{q}_h}[u] = O(h)$ by (14), making the second term $O(h^2)$ outright. This is the within-group term of (7).

Step 3: between-group term. By the C^2 Taylor expansion of μ,

$$\mu(y + hu) = \mu(y) + hJ(y)u + \tfrac{h^2}{2}Q(u) + O(h^3), \qquad Q(u)_a := \sum_{b,c} \partial^2_{y_b y_c}\mu_a(y)\, u_b u_c,$$

and since covariance is invariant to the additive constant $\mu(y)$,

$$\mathrm{Cov}_h[\mu(Y)] = \mathrm{Cov}_{\tilde{q}_h}\Big[hJ(y)u + \tfrac{h^2}{2}Q(u) + O(h^3)\Big] = h^2 J(y)\,\mathrm{Cov}_{\tilde{q}_h}[u]J(y)^\top + h^3\Big[J(y)\,\mathrm{Cov}_{\tilde{q}_h}(u, Q(u)) + \mathrm{sym}\Big] + h^4\,\mathrm{Cov}_{\tilde{q}_h}[Q \tag{17}$$

The leading term is $h^2 J(y)\big[I_k + O(h^2)\big]J(y)^\top = h^2 J(y)J(y)^\top + O(h^4)$ by (16). It remains to show the middle, apparently $O(h^3)$, cross term is in fact $O(h^4)$: this holds because $\mathrm{Cov}_{\tilde{q}_h}(u, Q(u))$ itself vanishes at $h = 0$ and is only $O(h)$, not $O(1)$. Indeed, since $Q(u)$ is a quadratic form in u, $\mathbb{E}_{\tilde{q}_h}[u\,Q(u)^\top]$ and $\mathbb{E}_{\tilde{q}_h}[u]\,\mathbb{E}_{\tilde{q}_h}[Q(u)]^\top$ both involve only the third moments $\mathbb{E}_{\tilde{q}_h}[u_a u_b u_c]$ of $\tilde{q}_h$ (for the second factor, through $\mathbb{E}_{\tilde{q}_h}[u] = O(h)$ paired with $\mathbb{E}_{\tilde{q}_h}[Q(u)] = O(1)$, which is itself an $O(h)$ quantity overall); expanding (13) to the relevant order,

$$\mathbb{E}_{\tilde{q}_h}[u_a u_b u_c] = \mathbb{E}_\varphi[u_a u_b u_c] + h\sum_e s_e(y)\,\mathbb{E}_\varphi[u_a u_b u_c u_e] + O(h^2) = 0 + h\big[s_c(y)\delta_{ab} + s_b(y)\delta_{ac} + s_a(y)\delta_{bc}\big] + O(h^2),$$

using $\mathbb{E}_\varphi[u_a u_b u_c] = 0$ and the fourth-moment identity above. Thus every third moment of $\tilde{q}_h$ is itself $O(h)$ (it is exactly 0 under the untilted Gaussian φ), so $\mathrm{Cov}_{\tilde{q}_h}(u, Q(u)) = O(h)$, and the cross term in (17) is $h^3 \cdot O(h) = O(h^4)$, absorbed into the stated $O(h^4)$. The final term $h^4\,\mathrm{Cov}_{\tilde{q}_h}[Q(u)] = O(h^4)$ trivially. Combining, $\mathrm{Cov}_h[\mu(Y)] = h^2 J(y)J(y)^\top + O(h^4)$, the between-group term of (7).

Substituting Steps 2–3 into (12) proves (11), i.e. the deterministic part of (7).

Step 4: Monte-Carlo fluctuation. Write $S_N := \frac{1}{N}\sum_j K_h(y - y^j)$ and $T_N := \frac{1}{N}\sum_j K_h(y - y^j)x^j$, so $\bar{x}_h(y) = T_N/S_N$. Each is an average of N i.i.d. terms; a standard kernel-density computation gives $\mathrm{Var}\big(K_h(y - Y)\big) = \Theta(h^{-k})$ (the kernel has height $\Theta(h^{-k})$ on an effective support of Lebesgue measure $\Theta(h^k)$, so its second moment is $\Theta(h^{-k}) \cdot \Theta(h^k) \cdot \Theta(h^{-k}) = \Theta(h^{-k})$, dominating the squared mean $\Theta(1)$), and likewise for $K_h(y - Y)X$. Hence $S_N = \mathbb{E}[K_h(y - Y)] + O_P\big(\sqrt{h^{-k}/N}\big)$ and $T_N = \mathbb{E}[K_h(y - Y)X] + O_P\big(\sqrt{h^{-k}/N}\big)$, i.e. each has *relative* fluctuation $O_P\big((Nh^k)^{-1/2}\big)$ around a mean of order $\Theta(1)$. A first-order Taylor (Slutsky) expansion of the ratio T_N/S_N around the population ratio $\mathbb{E}_h[X]$ then gives $\bar{x}_h(y) = \mathbb{E}_h[X] + O_P\big((Nh^k)^{-1/2}\big)$, exactly as in standard Nadaraya–Watson consistency theory; the same argument applied to the (also ratio-of-averages) quadratic functional $\mathrm{Cov}_h(y)$ gives $\mathrm{Cov}_h(y) = \mathrm{Cov}_h^{\mathrm{pop}}(y) + O_P\big((Nh^k)^{-1/2}\big)$, which is the remaining term of (7) and completes the proof. ☐

Proof of Proposition 11. Let X, X' be i.i.d. copies of $X_t \mid I = i \sim \mathcal{N}(tx^i, (1-t)^2 I_d)$ and $e(\cdot) = f_i(\cdot, t) - v(\cdot, t, y^i)$. Since $f_i(X, t) - f_i(X', t) = -(X - X')/(1-t)$ and $X - X' \sim \mathcal{N}(0, 2(1-t)^2 I_d)$,

$$\mathbb{E}|f_i(X,t) - f_i(X',t)|^2 = \frac{2(1-t)^2 d}{(1-t)^2} = 2d.$$

By L-Lipschitzness, $\mathbb{E}|v(X,t,y^i) - v(X',t,y^i)|^2 \le L^2\mathbb{E}|X - X'|^2 = 2dL^2(1-t)^2$. Also $\mathbb{E}|e(X) - e(X')|^2 = 2\mathbb{E}|e(X)|^2 - 2|\mathbb{E}e(X)|^2 \le 2\mathbb{E}|e(X)|^2$ (since X, X' are i.i.d.). Applying the $L^2(\Omega)$ triangle inequality to $f_i(X,t) - f_i(X',t) = [e(X) - e(X')] + [v(X,t,y^i) - v(X',t,y^i)]$,

$$\sqrt{2d} \;\le\; \sqrt{2\,\mathbb{E}|e(X)|^2} \;+\; \sqrt{2d}\,L(1-t),$$

hence $\sqrt{\mathbb{E}|e(X)|^2} \ge \sqrt{d}\,[1 - L(1-t)]_+$, i.e. $\mathbb{E}|e(X)|^2 \ge d[1 - L(1-t)]_+^2$, which is (10)'s pointwise bound with $e(X) = v(X_t, t, y^i) - f_i(X_t, t)$ up to sign. (Using the sharp identity $\mathbb{E}|e(X) - e(X')|^2 = 2\mathbb{E}|e(X)|^2 - 2|\mathbb{E}e(X)|^2 \le 2\mathbb{E}|e(X)|^2$, rather than the cruder bound $4\mathbb{E}|e(X)|^2$, is what preserves the constant d in (10); a weaker triangle-inequality step would give only a smaller floor.) ☐

Proof of Corollary 5. Substituting $s = 1 - t$, $\int_0^{1/L}(1 - Ls)^2\,ds = \big[s - Ls^2 + \frac{L^2 s^3}{3}\big]_0^{1/L} = \frac{1}{L} - \frac{1}{L} + \frac{1}{3L} = \frac{1}{3L}$. Proposition 11's bound (10)'s integrand holds for every training index i, hence for every $v \in \mathcal{F}_L$ and every $t \in [1 - 1/L, 1)$, so it survives averaging over $I \sim \mathrm{Unif}\{1, \dots, N\}$ and $t \sim \mathrm{Unif}(0,1)$, giving $\inf_{v \in \mathcal{F}_L} L_0(v) \ge d\int_{1-1/L}^{1}[1 - L(1-t)]^2\,dt = d/(3L)$. ☐